DAΦNE Collider with Crab Waist Scheme: from KLOE-2 to SIDDHARTA-2 Experiment

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DAΦNE Team

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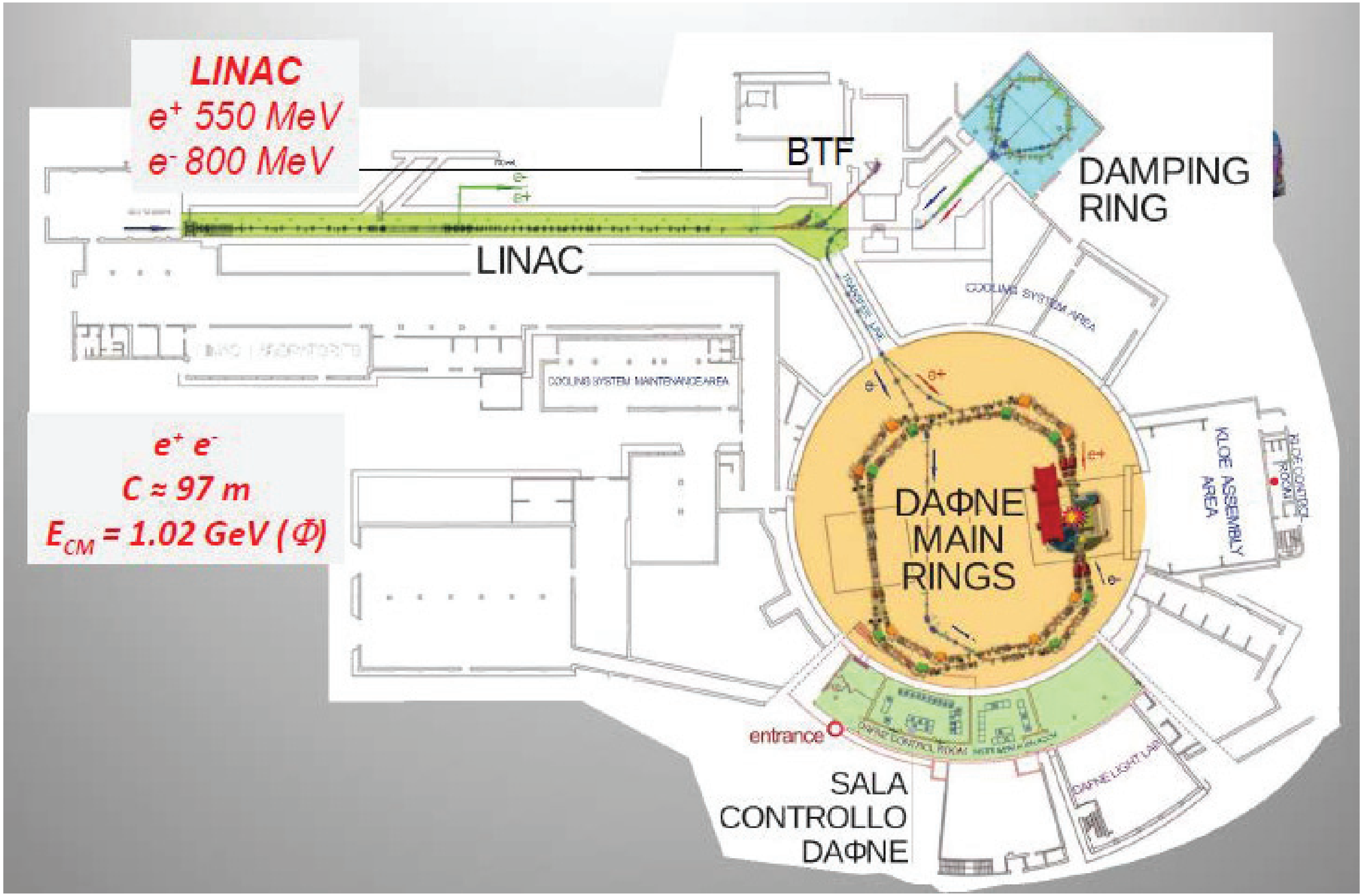
Colliders based on Crab Waist concept

Colliders	Location	Status
DAΦNE	Φ-Factory Frascati, Italy	In operation (SIDDHARTA, KLOE-2)
SuperKEKB	B-Factory Tsukuba, Japan	Phase II commissioning completed in 2018
SuperC-Tau	C-Tau-Factory Novosibirsk, Russia	Russian mega-science project
FCC-ee	Z,W,H,tt-Factory CERN,Switzerland	100 km, CDR at the end of 2018
CEPC	Higgs-Factory China	100 km, CDR released in September 2018
HIEPA	2-7 GeV China	Considered option

OUTLINE

- Introduction (DAΦNE Complex, Crab Waist Concept)
- Results and Limitations of the KLOE-2 Run
- Preparations for SIDDHARTA-2 Run
- Conclusions

DAΦNE Accelerator Complex



How to Increase Luminosity?

Luminosity

$$L = \frac{f_0}{4\pi} \frac{N^2}{\sigma_x \sigma_y}$$

Strength of Nonlinear Beam-Beam Interaction

$$\xi_y = \frac{r_e \beta_y}{2\pi\gamma} \frac{N}{\sigma_y \sigma_x}$$

Nonlinear resonances
Working points' spread
Chaos........................

Beam size blow up,
Lifetime reduction,
Instabilities

<u>The principal idea is:</u>

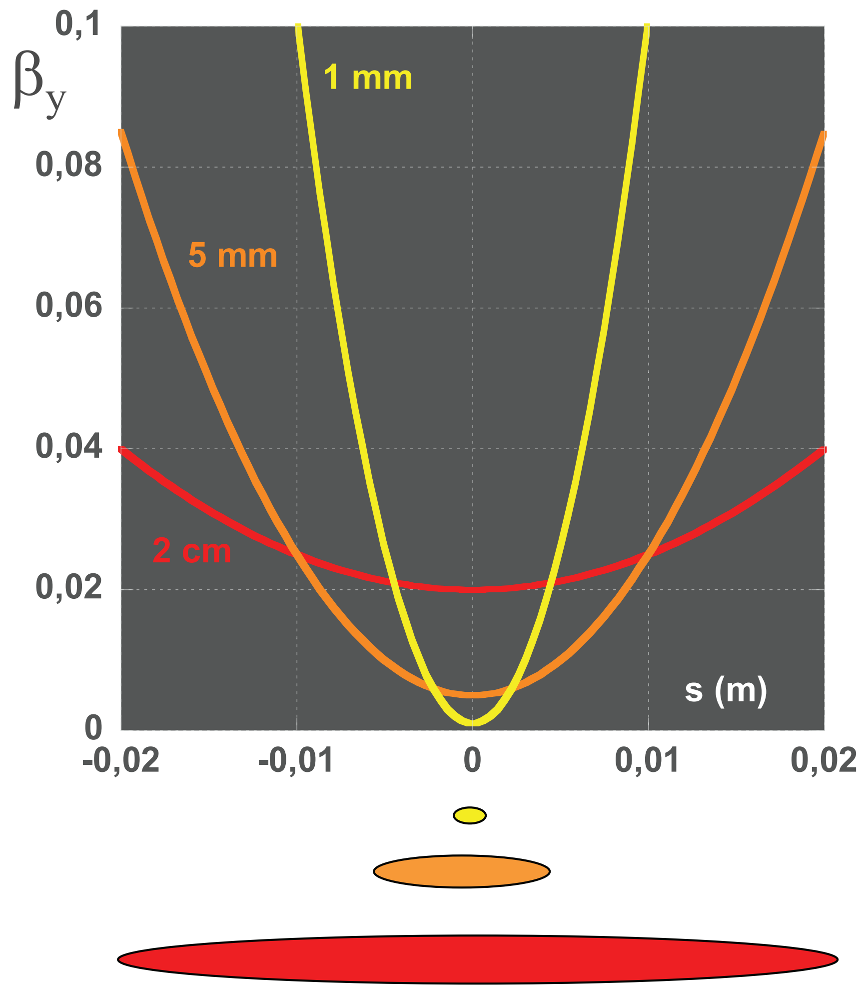
1) To linearize the nonlinear interaction

2) To weaken the effects of the nonlinear interaction

$$\sigma_y = \sqrt{\varepsilon_y \beta_y}$$

The "hourglass" effect

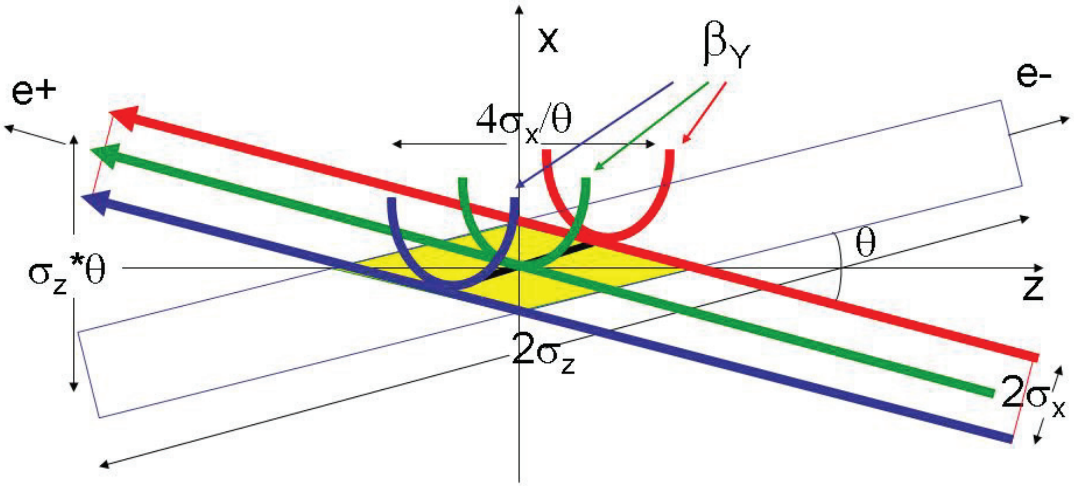
However, β_y^* cannot be shorter than the bunch length, because of the "hourglass effect"



Bunch distribution

Crab Waist in 3 Steps

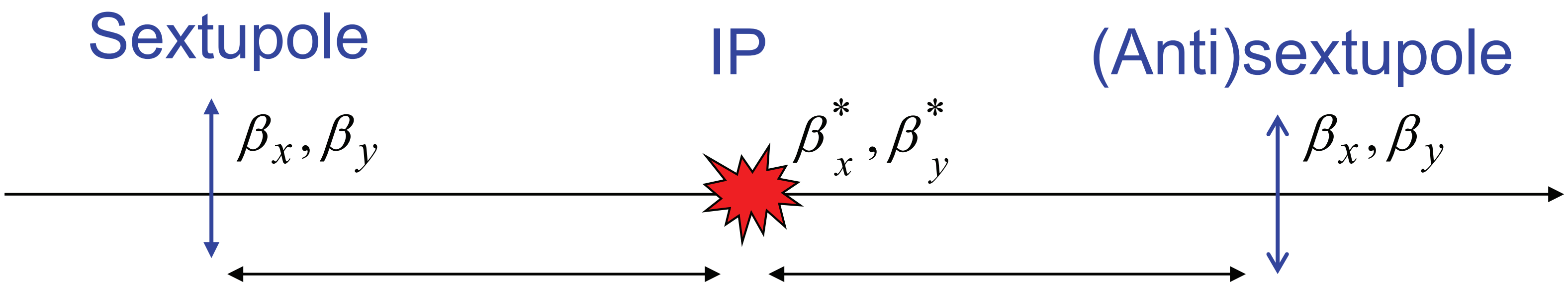
1. Large Piwinski's angle $\Phi = tg(\theta/2)\sigma_z/\sigma_x$
2. Vertical beta comparable with overlap area $\beta_y \approx 2\sigma_x/\theta$
3. Crab waist transformation $y = xy'/\theta$



1. *P.Raimondi, 2° SuperB Workshop, March 2006*

2. *P.Raimondi, D.Shatilov, M.Zobov, physics/0702033*

Crabbed Waist Scheme



$$\Delta\mu_y = \frac{\pi}{2} \qquad \Delta\mu_y = \frac{\pi}{2}$$

$$\Delta\mu_x = \pi \qquad \Delta\mu_x = \pi$$

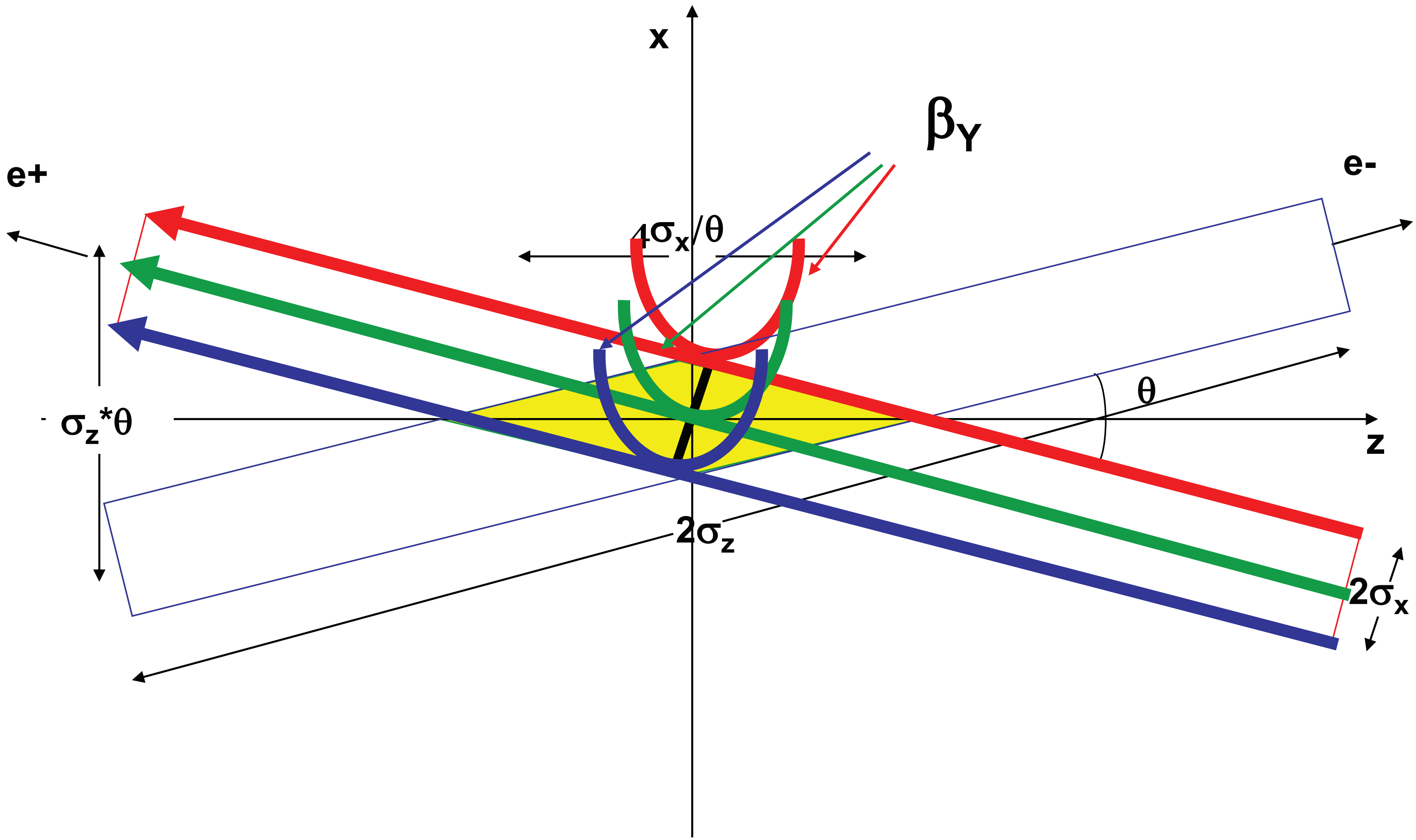
Sextupole strength

$$K = \frac{1}{2\theta} \frac{1}{\beta^*_y \beta_y} \sqrt{\frac{\beta^*_x}{\beta_x}}$$

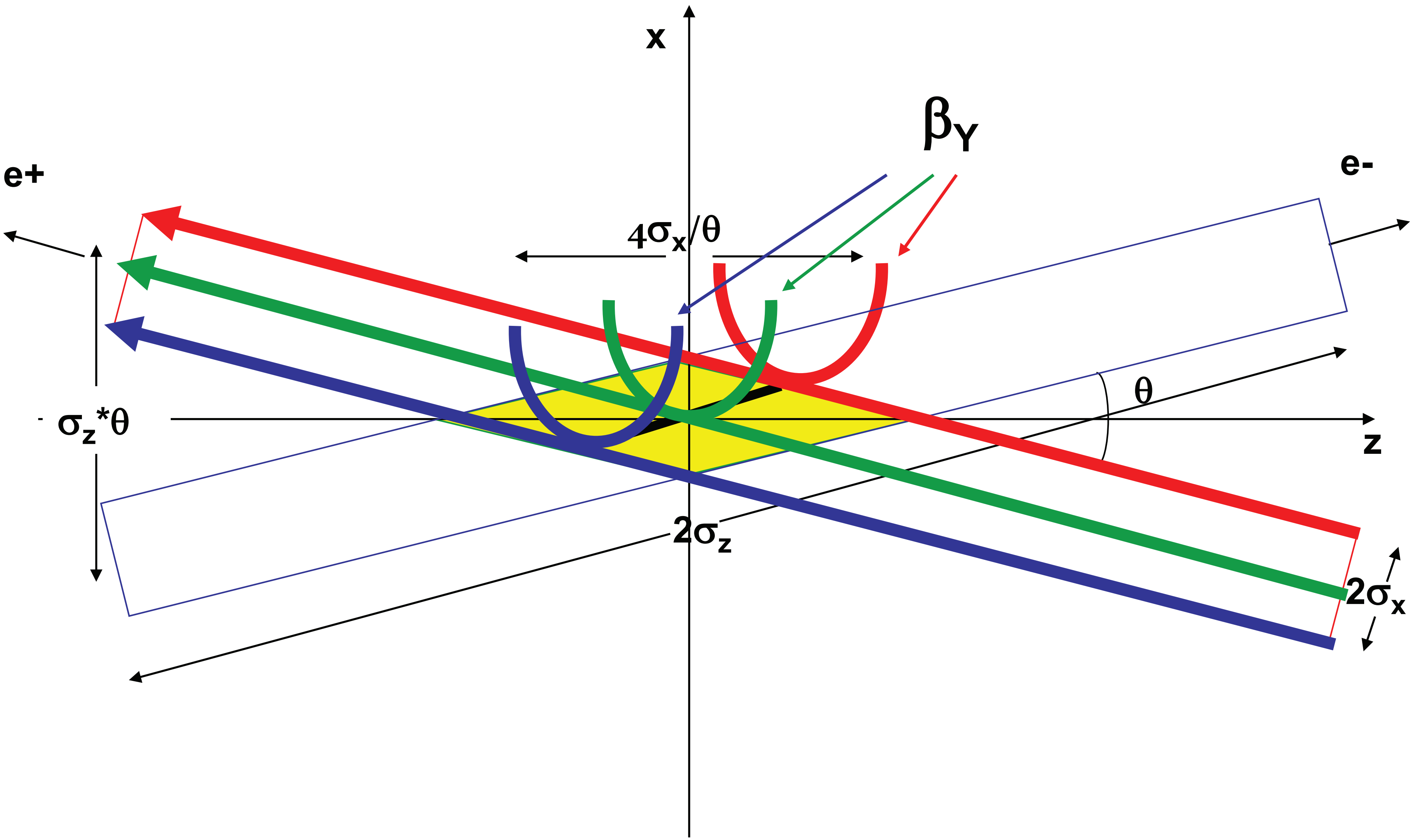
Equivalent Hamiltonian

$$H = H_0 + \frac{1}{2\theta} x p_y^2$$

$$\boxed{\beta_y = \beta^*_y + \frac{(s - x/\theta)^2}{\beta^*_y}}$$



x
β_Y
e+
e-
$4\sigma_x/\theta$
θ
$\sigma_z{*}\theta$
z
$2\sigma_z$
$2\sigma_x$


x
β_Y
e+
e-
$4\sigma_x/\theta$
θ
$\sigma_z*\theta$
z
$2\sigma_z$
$2\sigma_x$

Crabbed Waist Advantages

1. Large Piwinski's angle

$$\Phi = tg(\theta/2)\sigma_z/\sigma_x$$

- a) Luminosity gain with N
- b) Very low horizontal tune shift
- c) Vertical tune shift decreases with oscillation amplitude

2. Vertical beta comparable with overlap area

$$\beta_y \approx 2\sigma_x/\theta$$

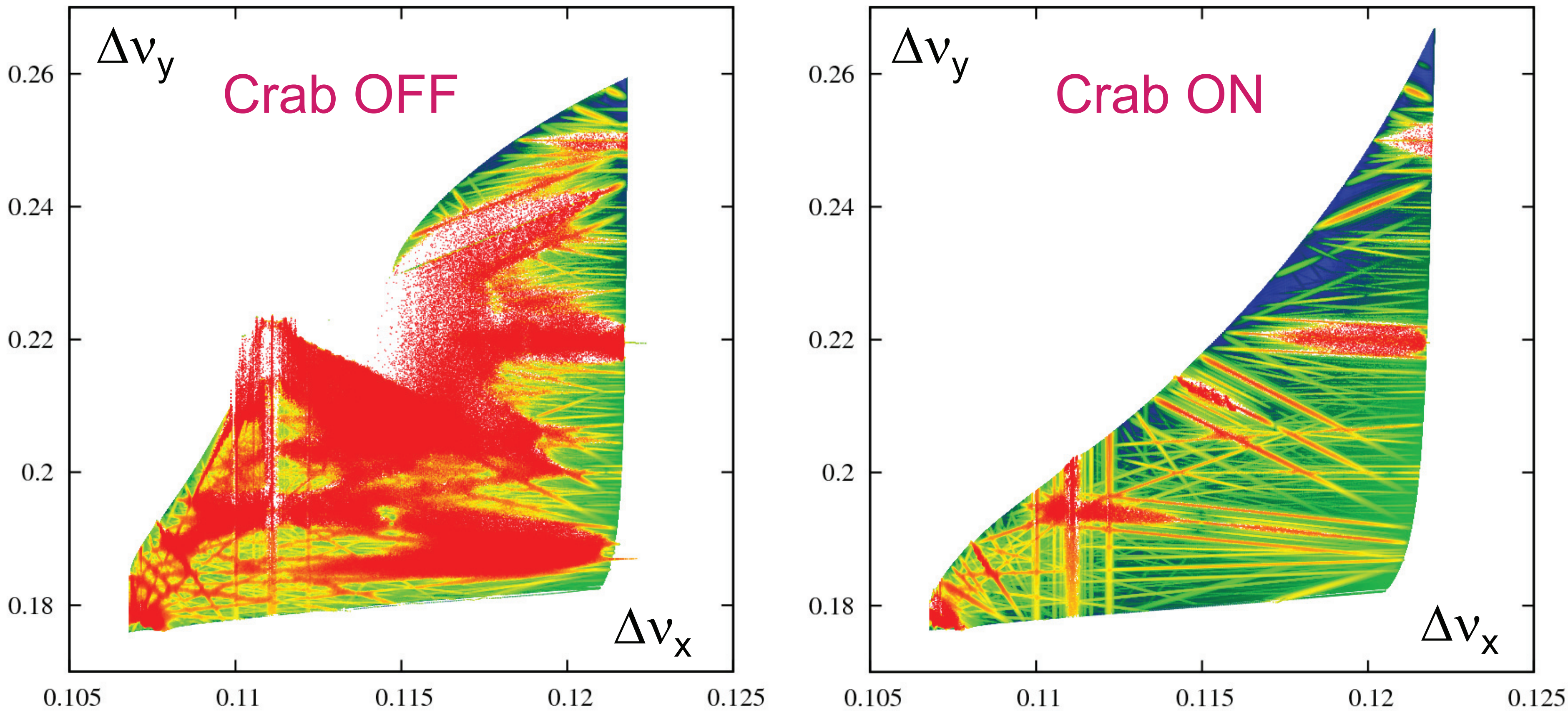
- a) Geometric luminosity gain
- b) Lower vertical tune shift
- c) Suppression of vertical synchro-betatron resonances

3. Crabbed waist transformation

$$y = xy'/\theta$$

- a) Geometric luminosity gain
- b) Suppression of X-Y betatron and synchro-betatron resonances

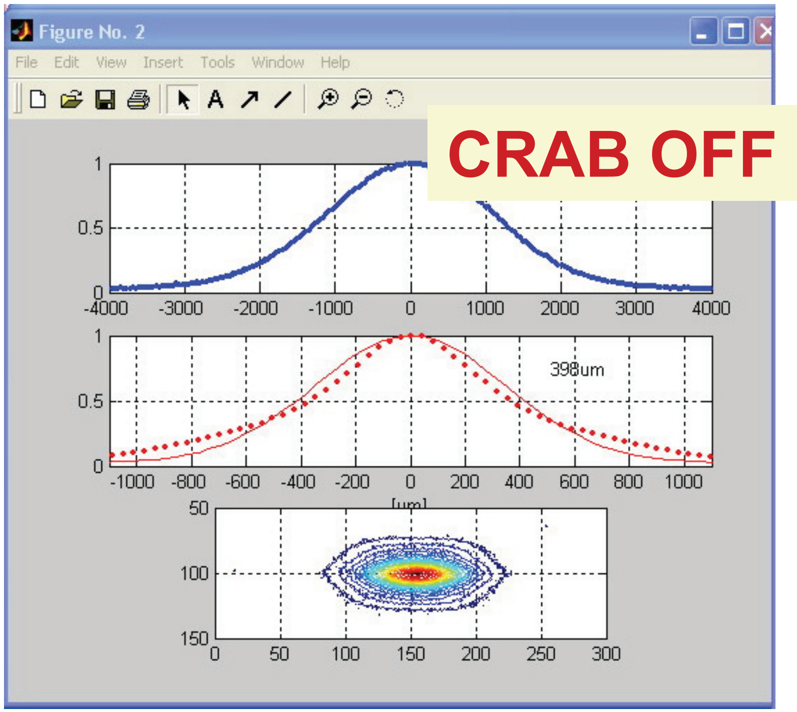
Frequency Map Analysis of Beam-Beam Interaction

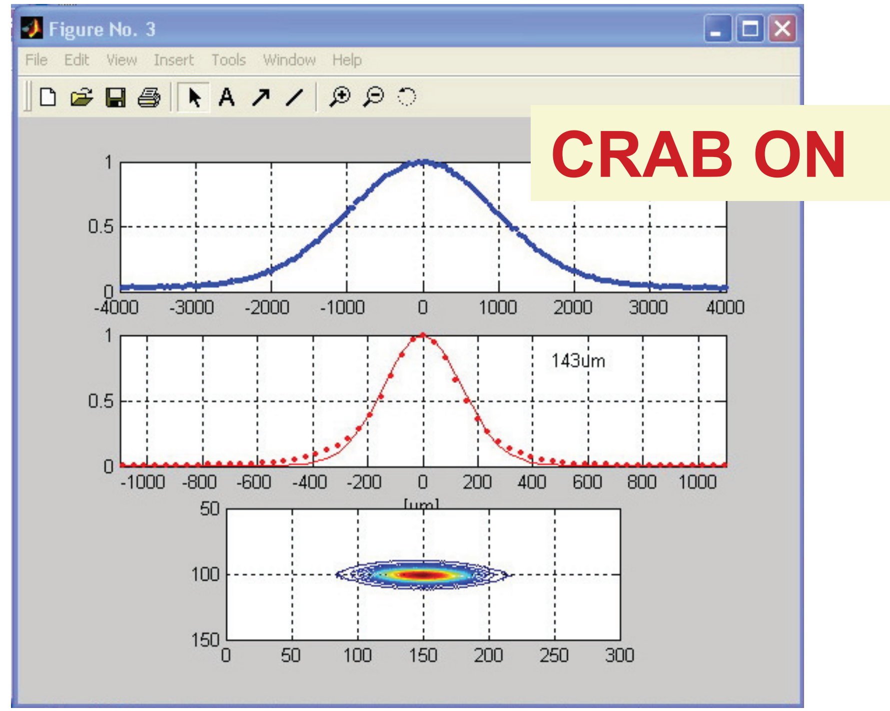


D.Shatilov, E.Levichev, E.Simonov and M.Zobov

Phys.Rev.ST Accel.Beams 14 (2011) 014001

Effect of CW Sextupoles

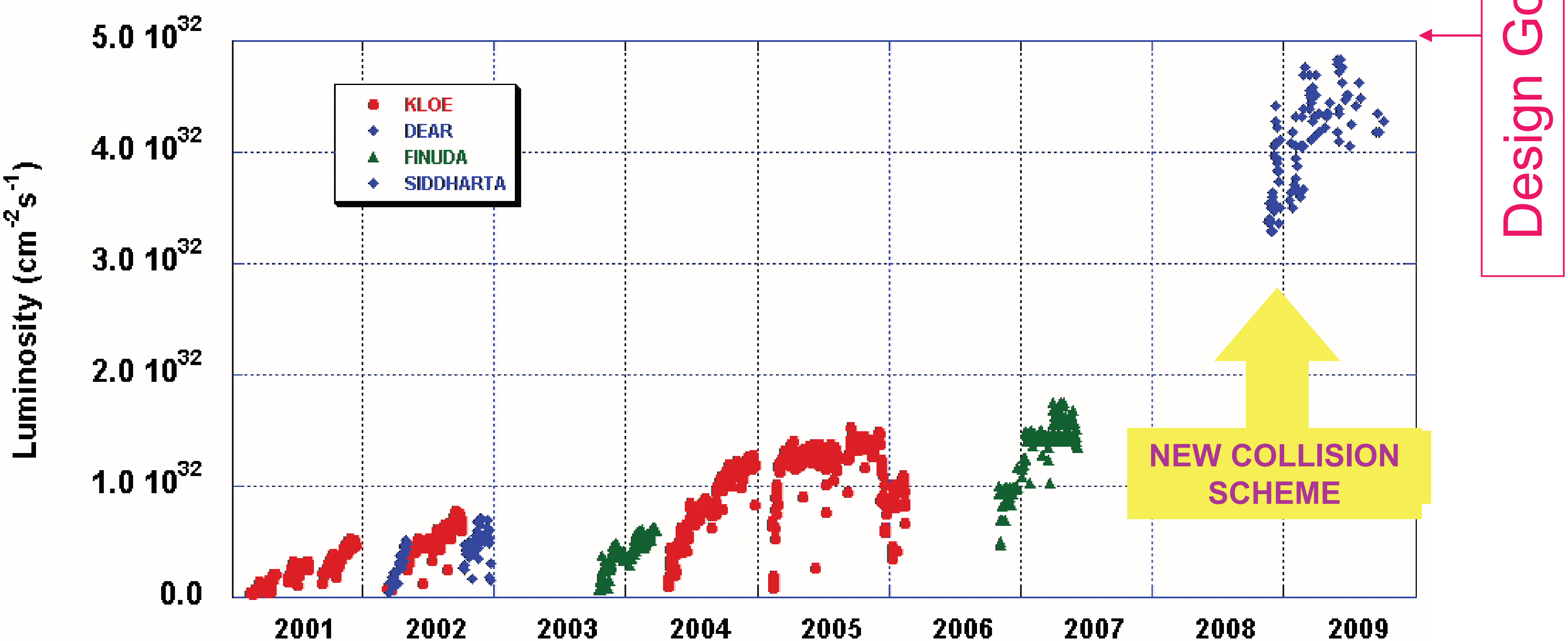


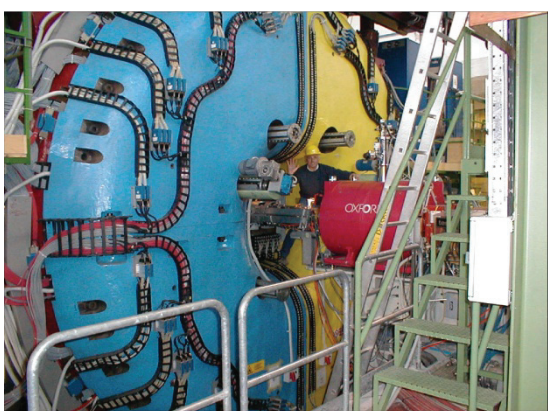


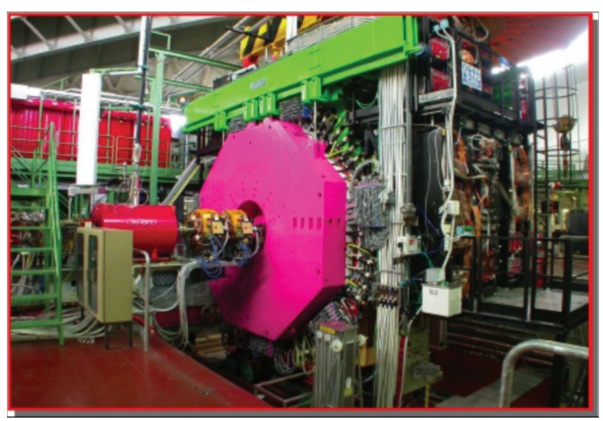
Switching on the CW sextupoles

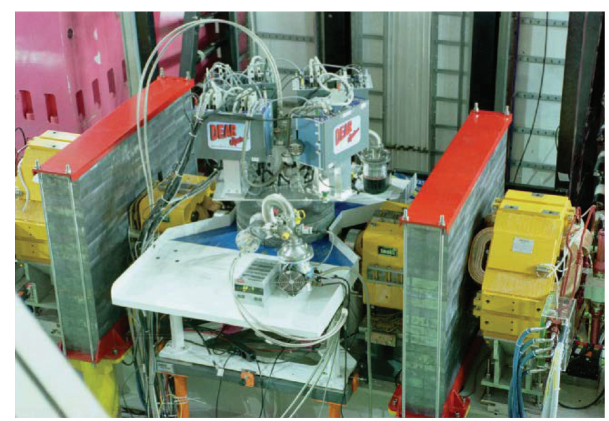
1. Beam sizes shrink
2. Beam-beam tails disappear
3. Luminosity and Lifetime are improved
4. Detector background gets lower

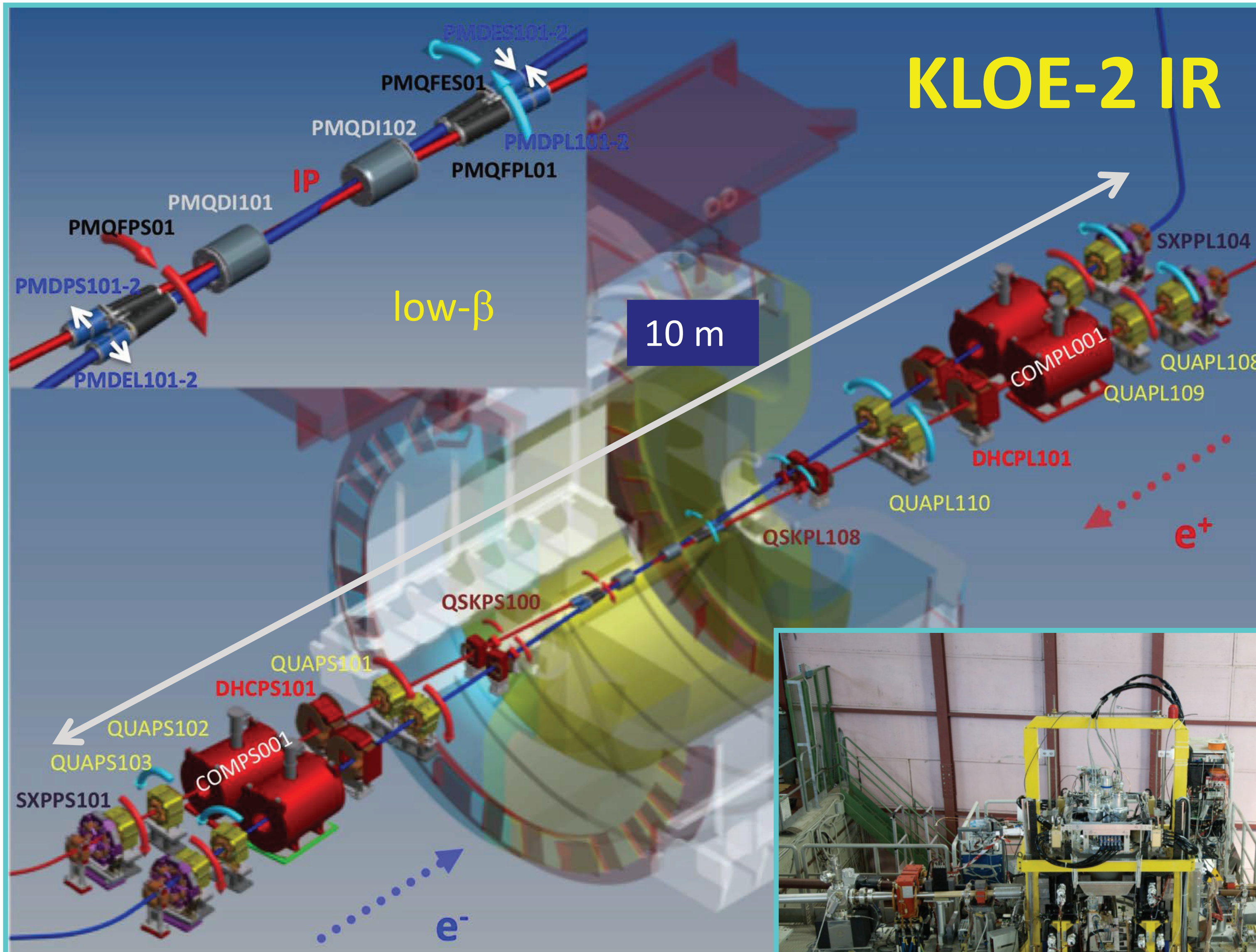
DAΦNE Peak Luminosity











C. Milardi *et al 2012 JINST 7 T03002.*

Coupling Correction

•$\int_{KLOE}$*B•dl canceled by 2 anti-solenoids for each beam*

$$\int_{KLOE} B \cdot dl = 2.048 \quad [Tm] \quad \rightarrow \quad I_{KLOE} = 2300[A]$$

$$\int_{comp} B \cdot dl = \pm 1.024 \quad [Tm] \quad \rightarrow \quad I_{comp} = 867[A]$$

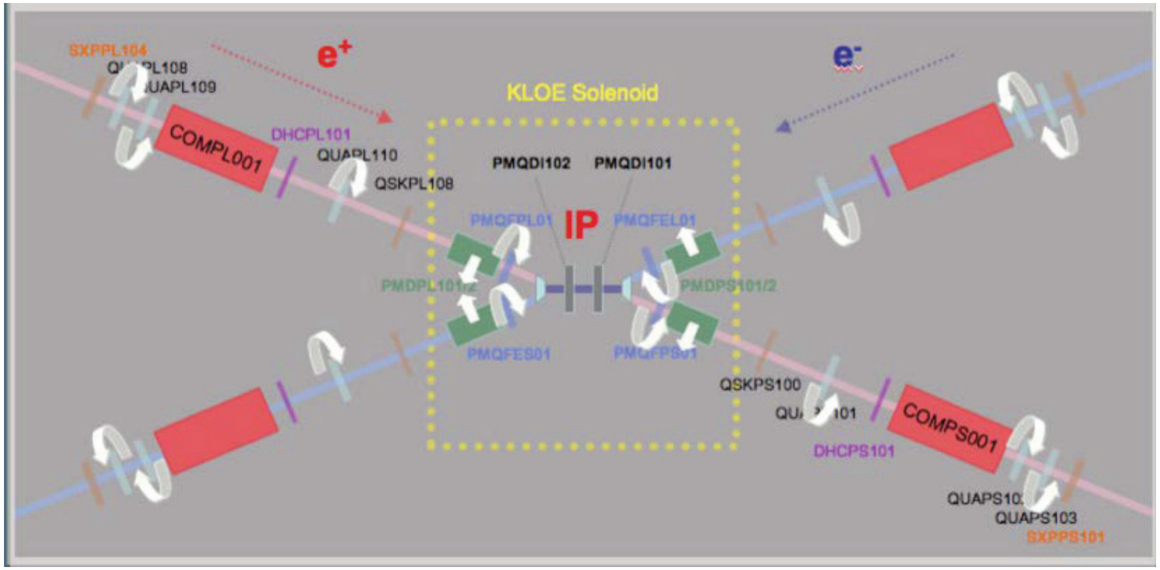
In order to have coupling compensation also for off-energy particles

	Z from the IP [m]	Quadrupole rotation angles [deg] *Anti-solenoid current [A]*
PMQDI101	0.415	**0.0**
PMQFPS01	0.963	**-4.48**
QSKPS100	2.634	**used for fine tuning**
QUAPS101	4.438	**-13.73**
QUAPS102	8.219	**0.906**
QUAPS103	8.981	**-0.906**
COMPS001	*6.963*	*72.48 (optimal value 86.7)*

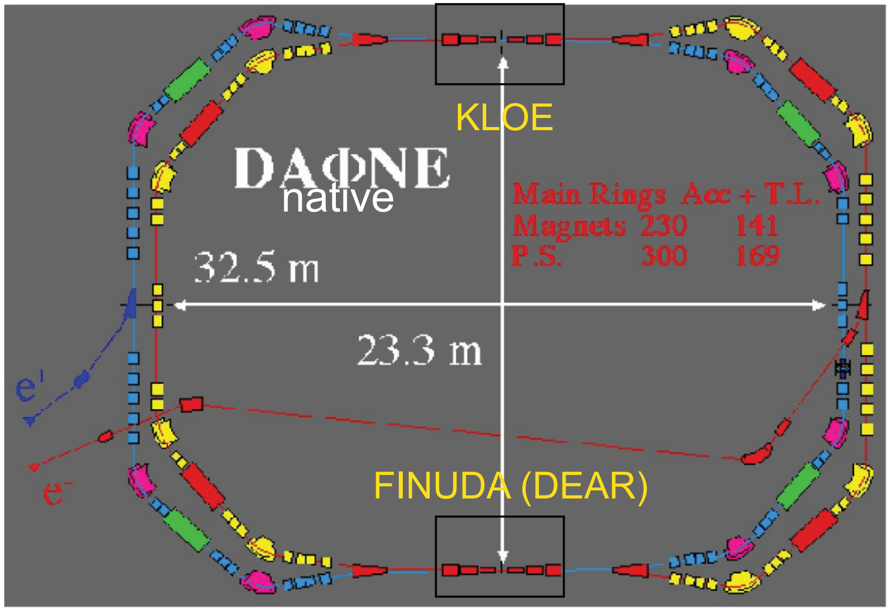
Fixed QUAD rotations

K *is expected to be lower than for KLOE past*

$K_{KLOE1} = 0.2 \div 0.3$ %



DAΦNE Layout and Parameters

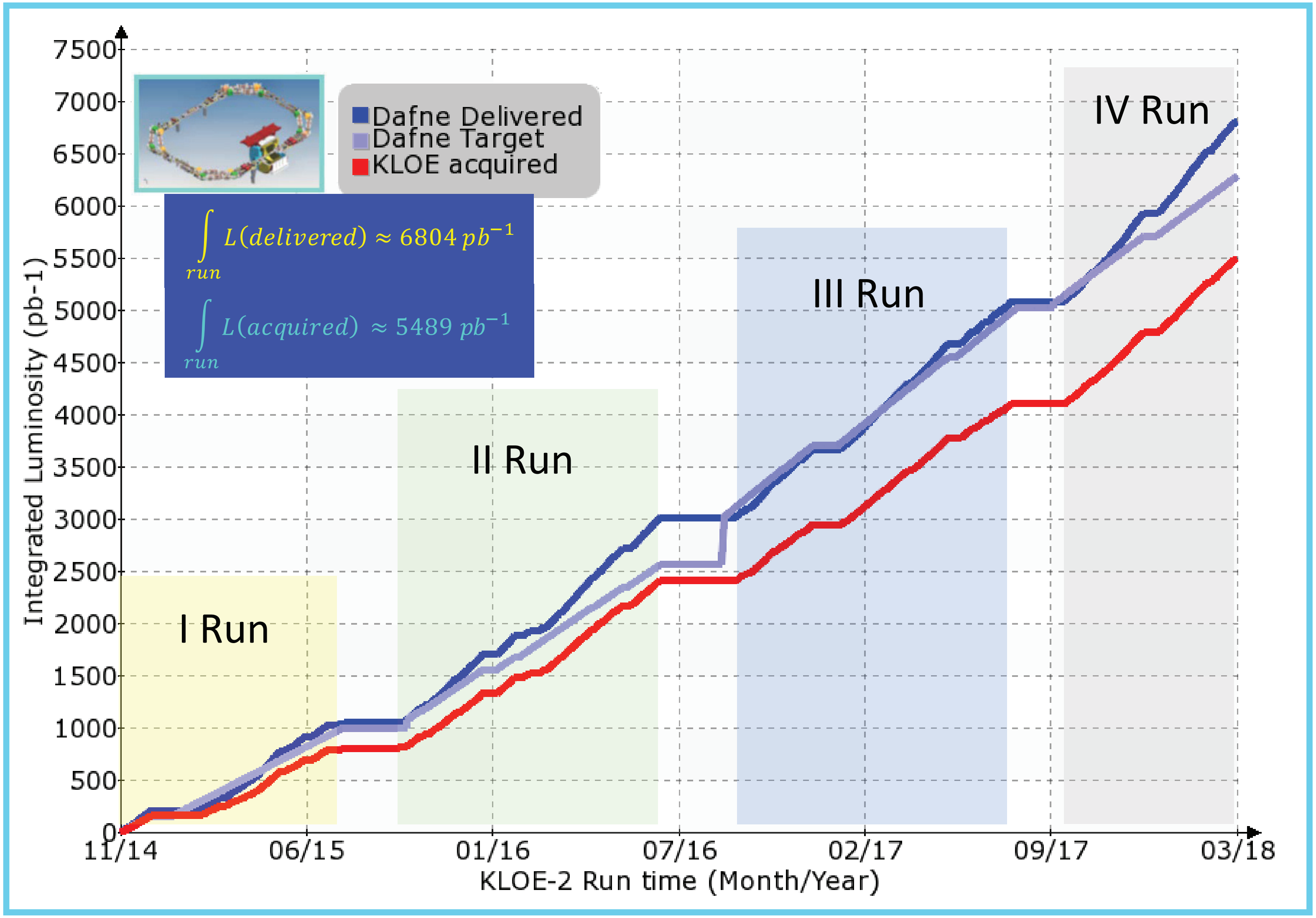


"Proposal for a Φ-factory", LNF-90/031 (IR),1990.

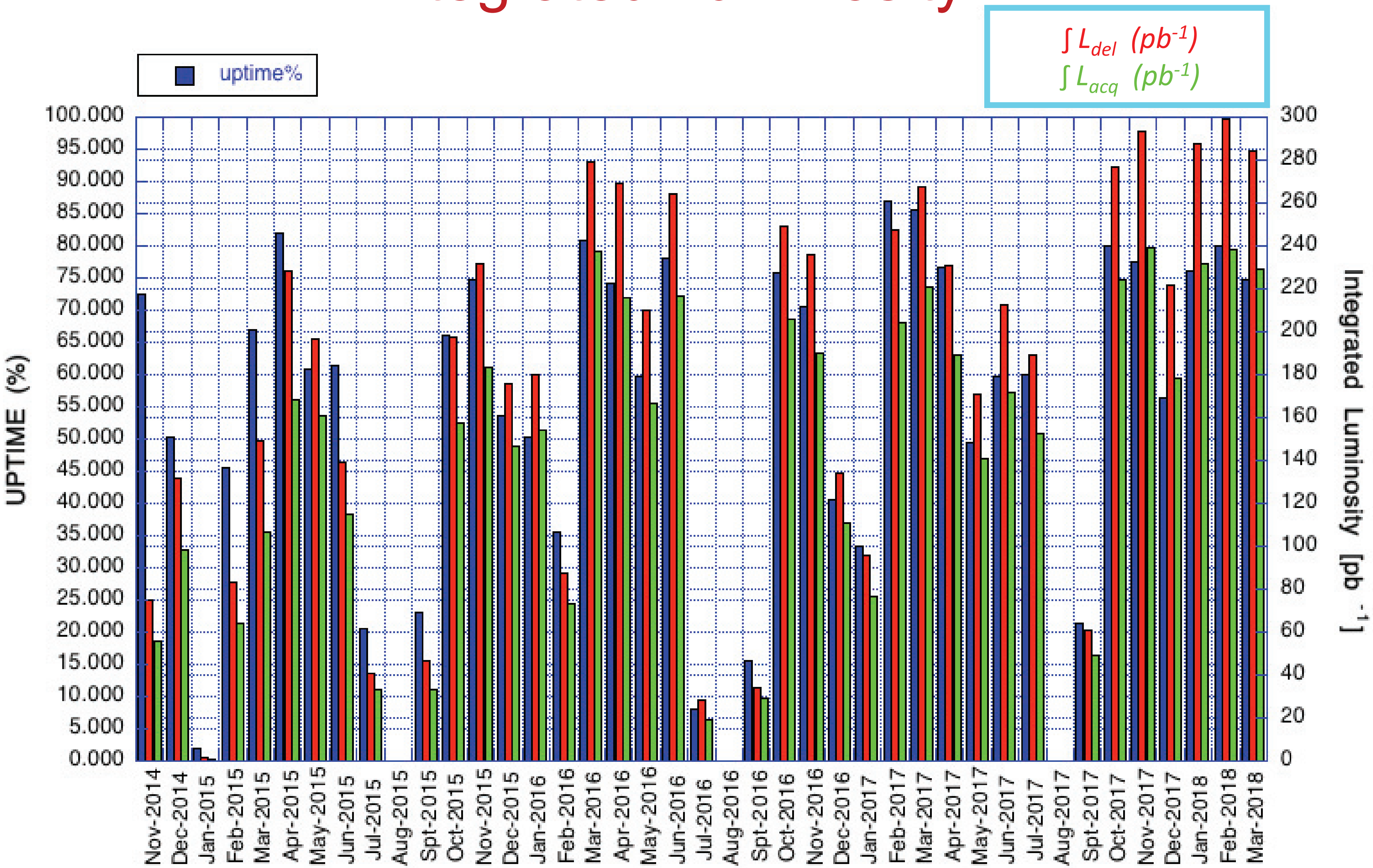
	DAΦNE native	DAΦNE Crab-Waist
Energy (MeV)	510	510
$\theta_{cross}/2$ (mrad)	12.5	25
ε_x (mm•mrad)	0.34	0.28
β_x^* (cm)	160	23
σ_x^* (mm)	0.70	0.25
$\Phi_{Piwinski}$	0.6	1.5
β_y^* (cm)	1.80	0.85
σ_y^* (μm) low current	5.4	3.1
Coupling, %	0.5	0.5
Bunch spacing (ns)	2.7	2.7
I_{bunch} (mA)	13	13
σ_z (mm)	25	15
N_b	120	120

Colliding Beams have:
- *low E*
- *high currents*
- *short bunch spacing 2.7 nsec*
- *long damping time*

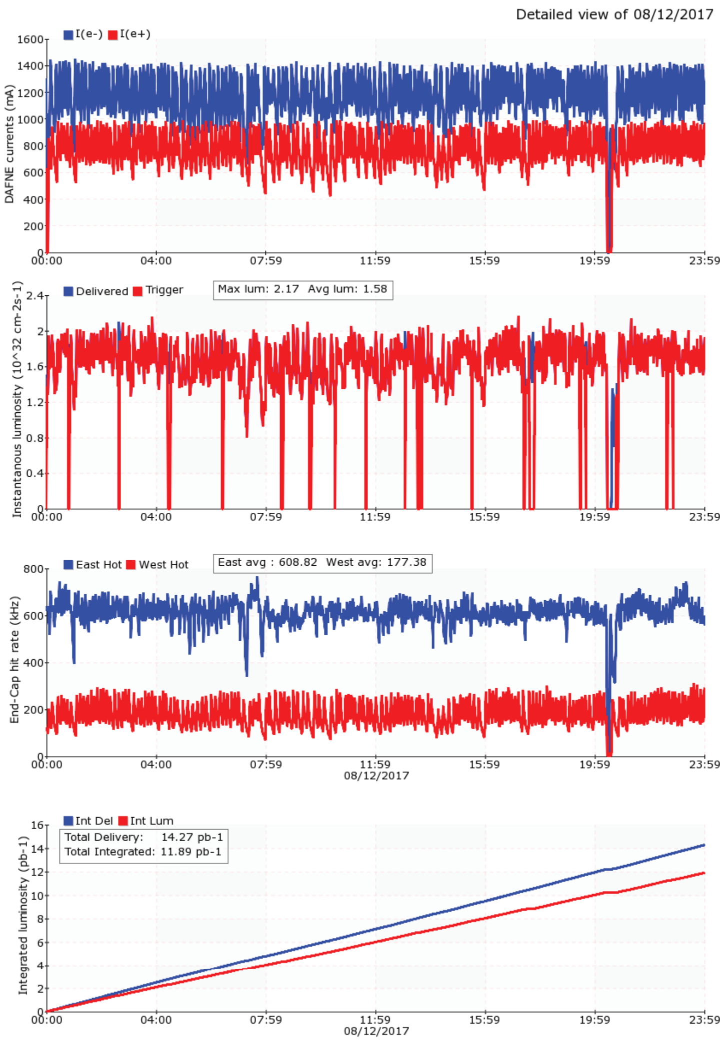
KLOE-2 Run Overview



DAΦNE Monthly Uptime and Integrated Luminosity



Highest Daily Integrated Luminosity

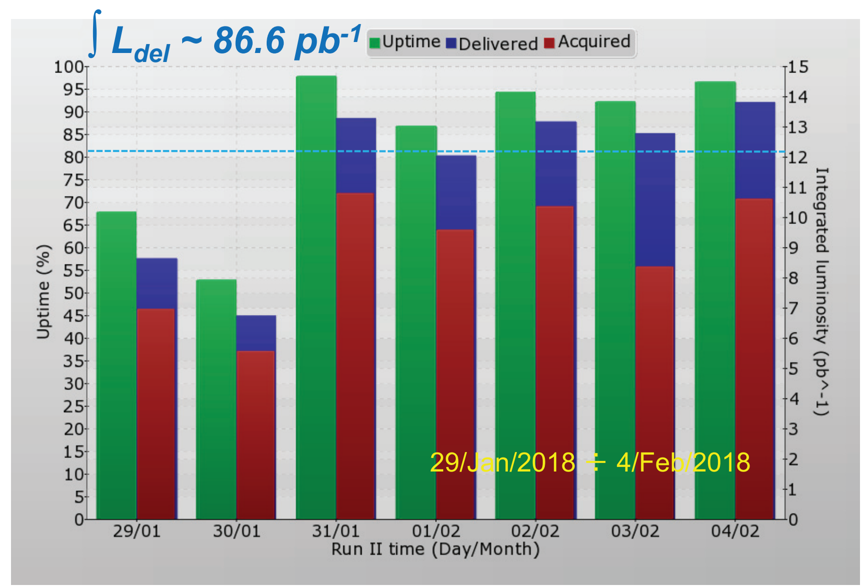


$\int L_{del} \sim 14.3\ pb^{-1}$

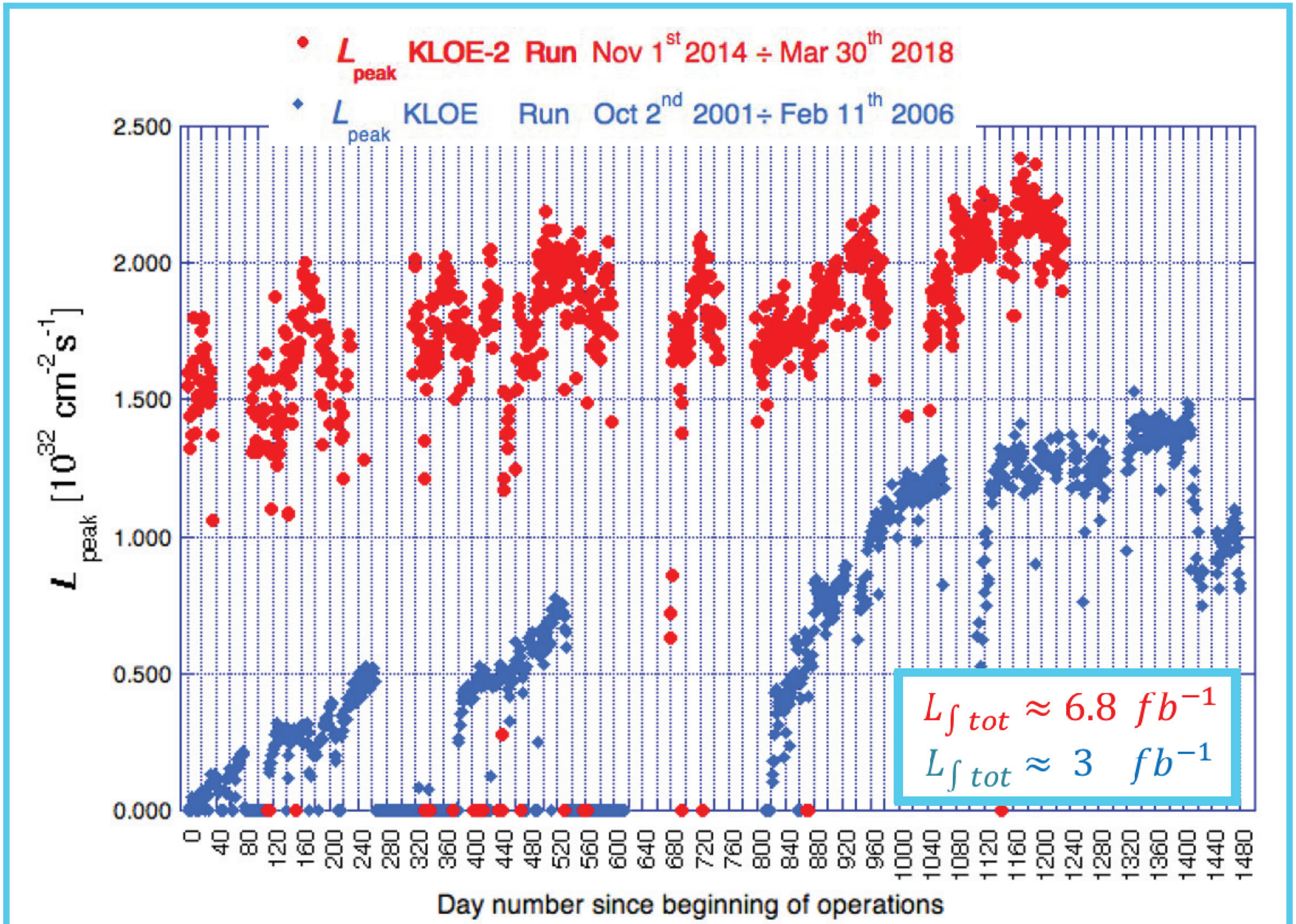
$\int L_{acq} \sim 11.9\ pb^{-1}$

Uptime ~98%

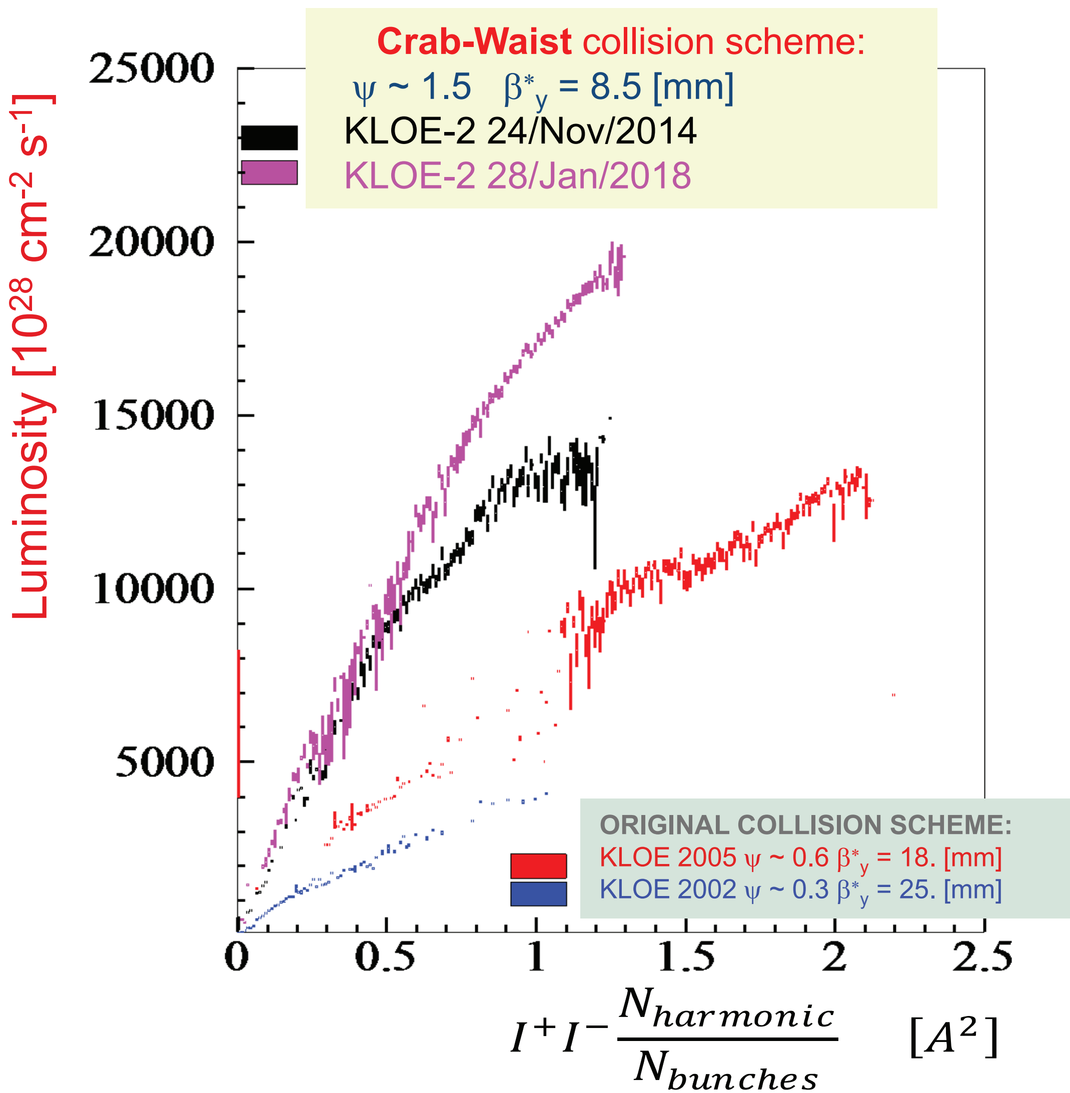
- 106 bunches
- Sustainable background



Crab Waist Peak Luminosity Gain



Specific Luminosity Gain



Main Issues to Take Care of

- More strict background control
- Higher beam coupling impedance
- e-cloud effects and instabilities

Managing Background

Background Challenges (with respect to KLOE)

1. Higher background due to lower emittances (Touschek effect)
2. Smaller beam pipe radius (from 44 mm to 27.5 mm)
3. New detector layers closer to the interacton point

Backdroung Mitigation

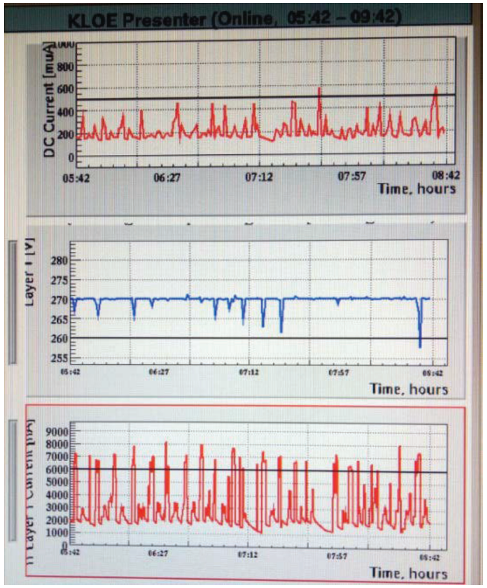
1. Change of the electron ring working point
2. Nonlinear dynamics optimization
3. RF frequency change to improve the energy acceptance
4. Lowering the RF voltage
5. Optimized collimator settings

Background Control

The new detector layers installed around the beam pipe posed new tight requirements on background level and control.

Criteria for acceptable background became:

- counting rate on the detector endcaps
- current amplitude measured by the different drift chamber sectors
- discharge threshold on the innermost IT layer



Background on the IT was heavily dependent on the injection process which had to be accurately optimized and stabilized

Even small drifts in the energy of the incoming beam, 0.01 ÷ 0.02 %, were causing unaffordable background level.

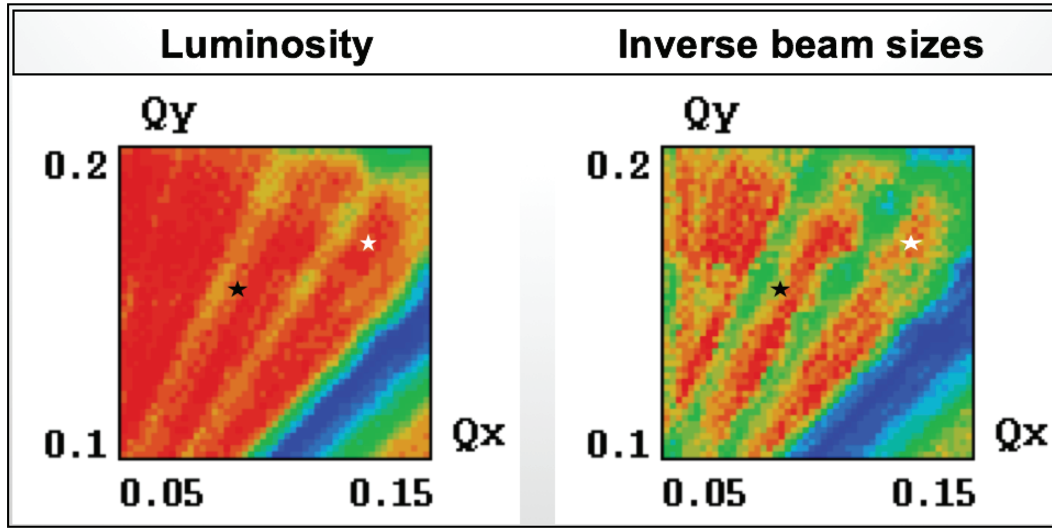
e^- Ring Working Point Scan

Numerical simulations aiming at improving:

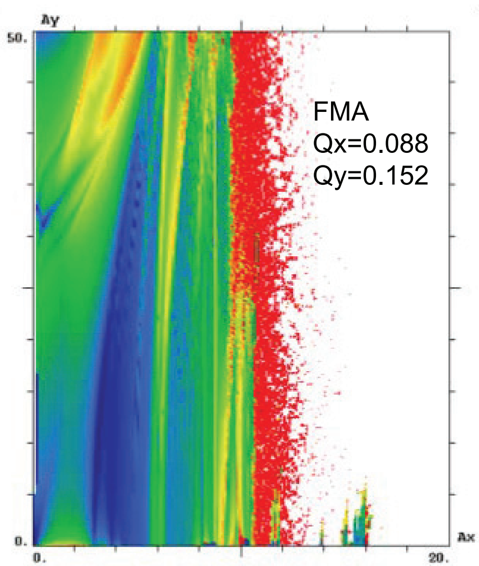
- dynamical aperture
- specific luminosity

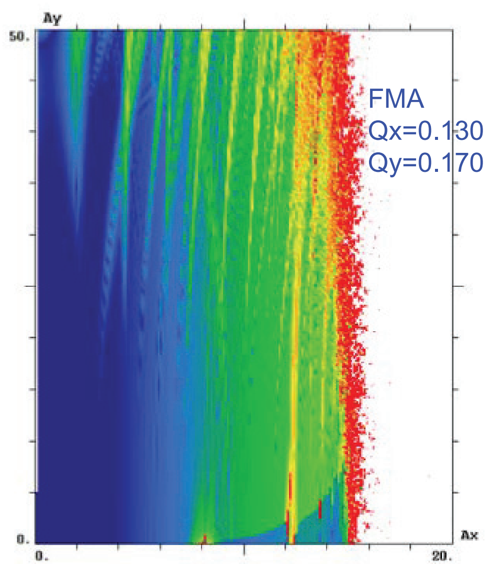
led to define a new working point for the e^- ring

Qx=0.135 Qy=0.17



M. Zobov *et al.*, *IEEE Trans. Nucl. Sci.,* Vol. 63, no. 2, pp. 818-822,2016.

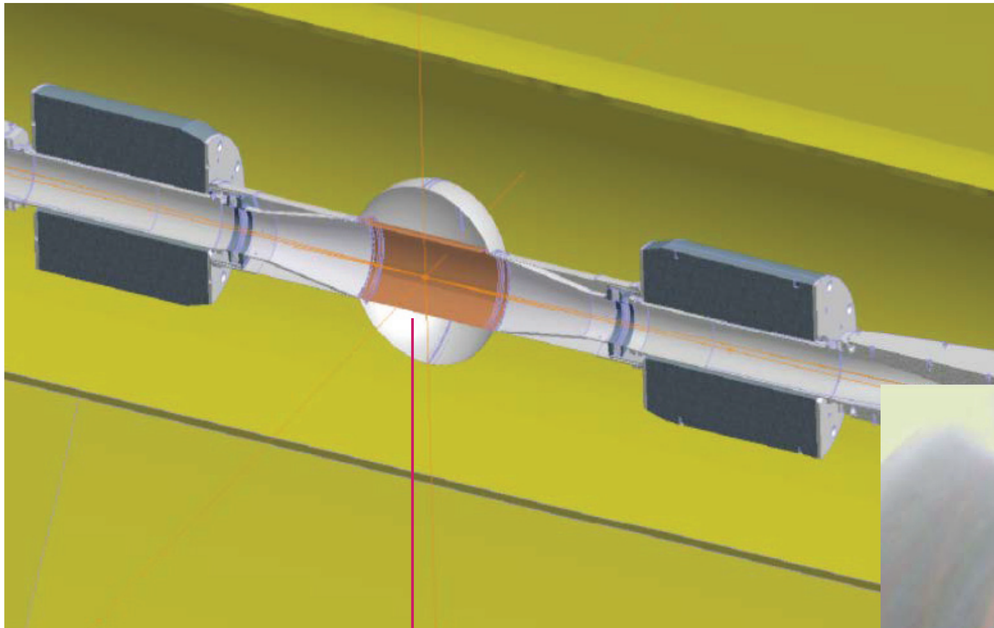




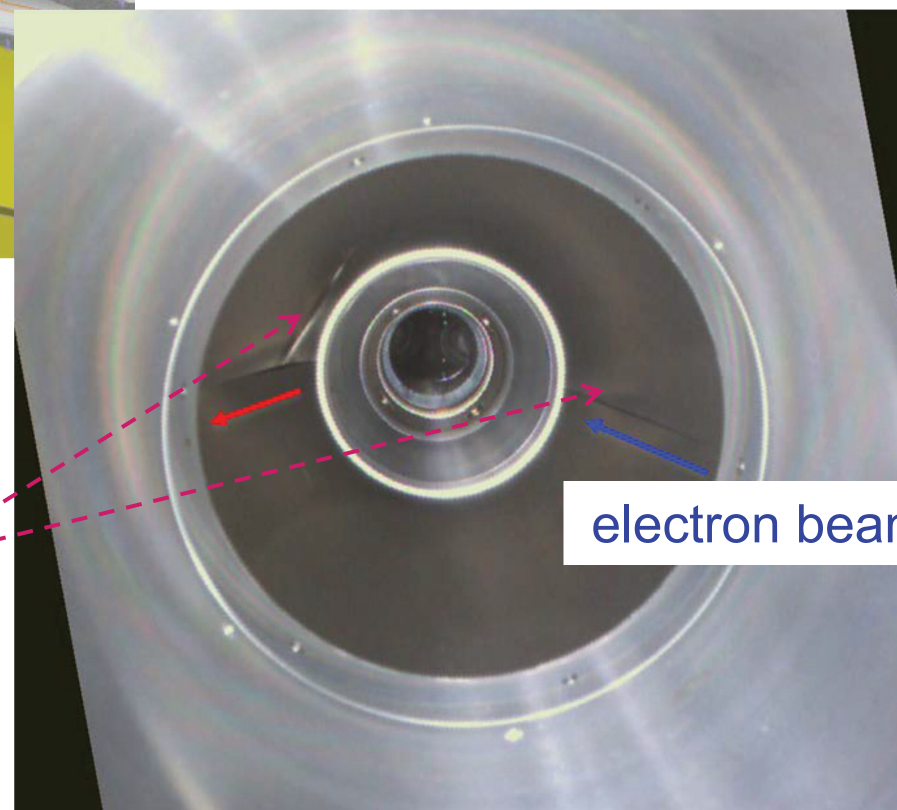
achieving:

- larger DA 2 – 3 σ
- Improved injection efficiency
- higher beam lifetime
- reduced background ~ 20%
- higher luminosity ~ 7%

IR Vacuum Chamber Inspection



View from the IR branch side
downstream the positron beam



Beryllium screen damages are closer to the electron beam orbit

Impedance Related Effects and Cures

Higher Impedance

1. New interaction region
2. Additional dump kickers
3. e-Cloud clearing electodes
4. Narrower collimator vacuum chamber
5. etc..

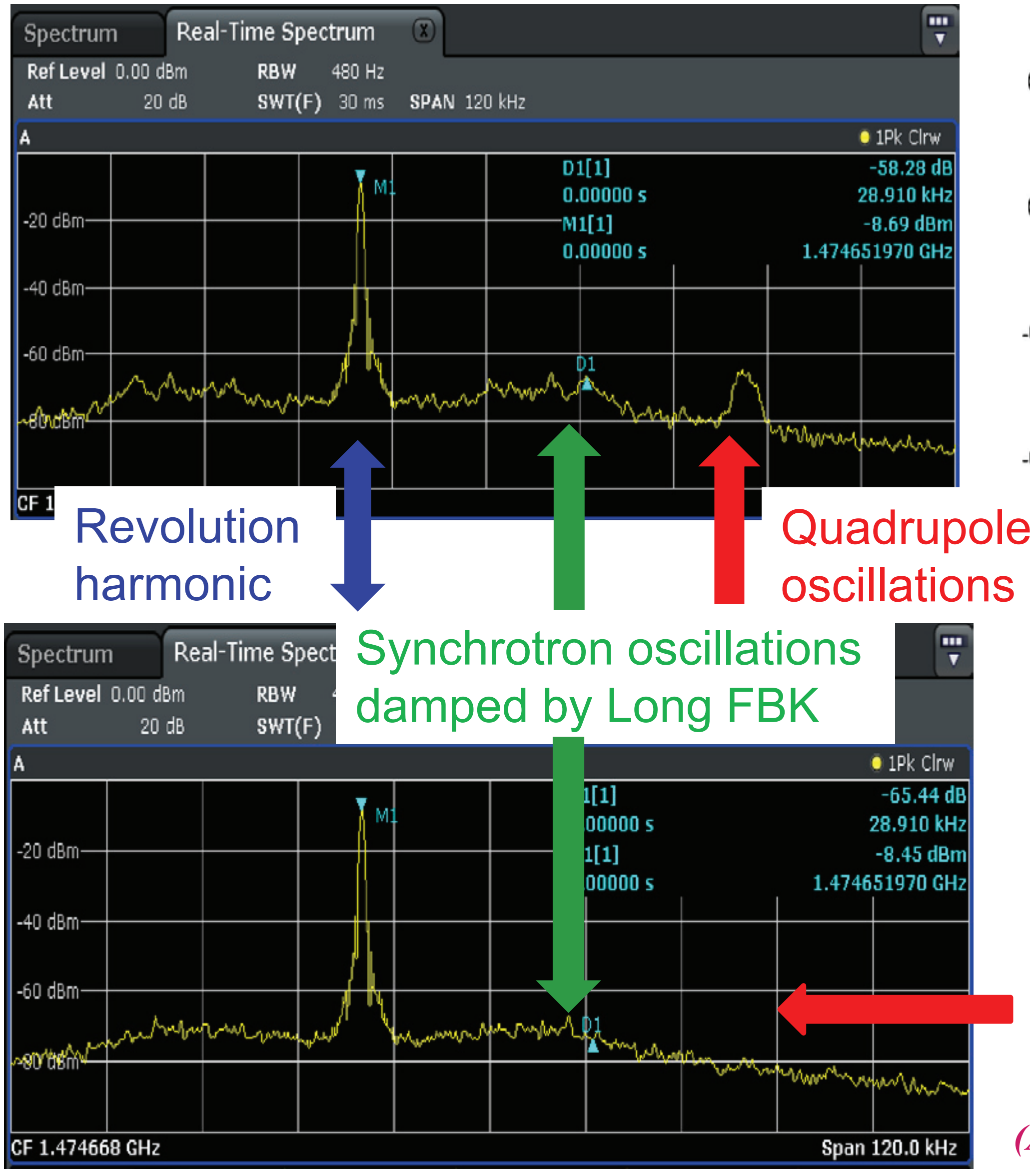
Lower Thresholds of

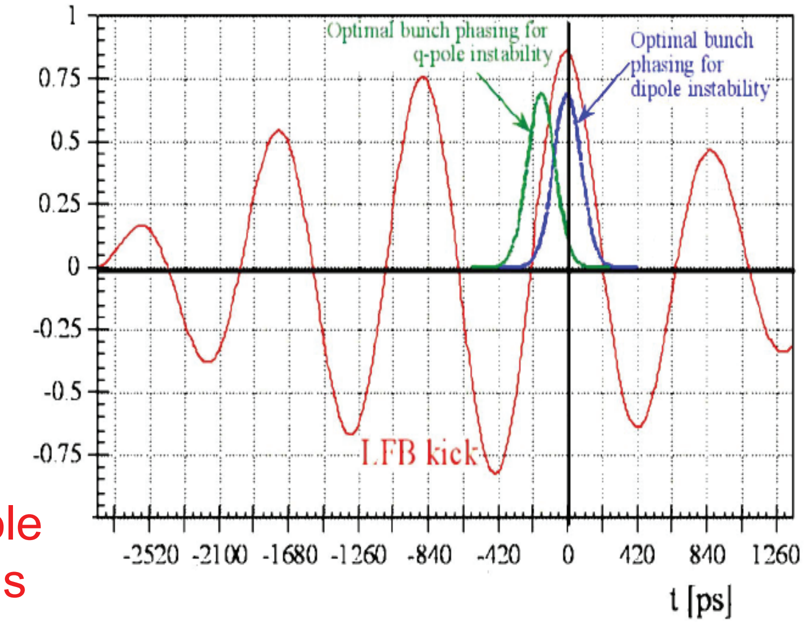
1. Longitudinal bunch shape oscillations
2. Transverse beam size increase

Cures

1. Special technique of using the dipole feedback system to suppress the quadrupole bunch shape oscillations
2. Lower RF voltage

Longitudinal Quadrupole Oscillations





The quadrupole oscillations result in injection saturation

The instability is cured by providing different kicks for bunch «tails» and «heads»

(A.Drago et al., PRST-AB, 6, 052801-1-11, 2003)

e-Cloud in DAΦNE

Challenges

1. Aluminium vacuum chamber
2. Bunch separation of 2.7 ns
3. High beam current

Mitigation

1. Feedback systems
2. Solenoids in straight sections
3. Clearing electrodes
4. Lower RF voltage
5. others

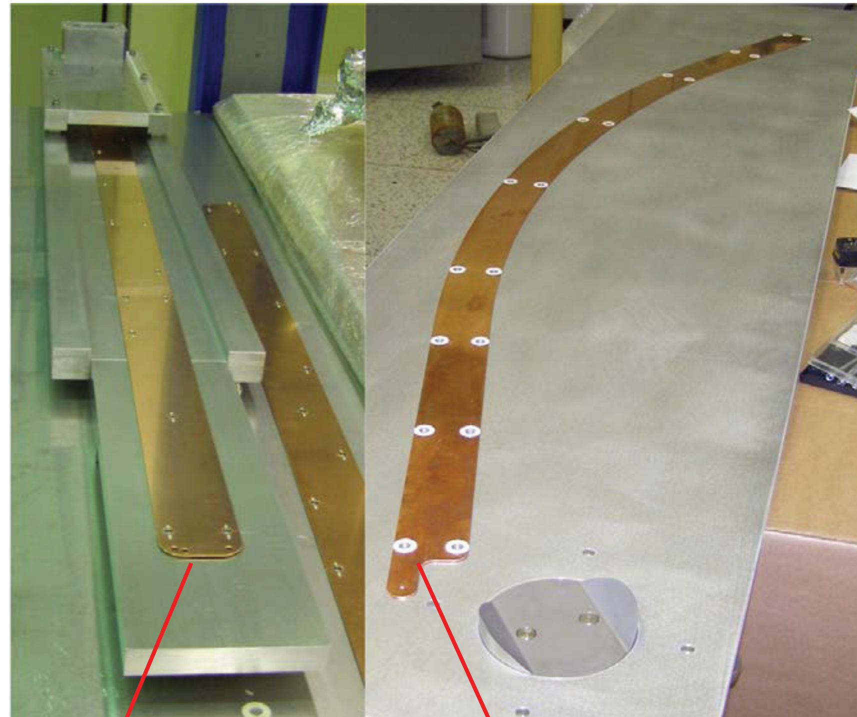
Effects

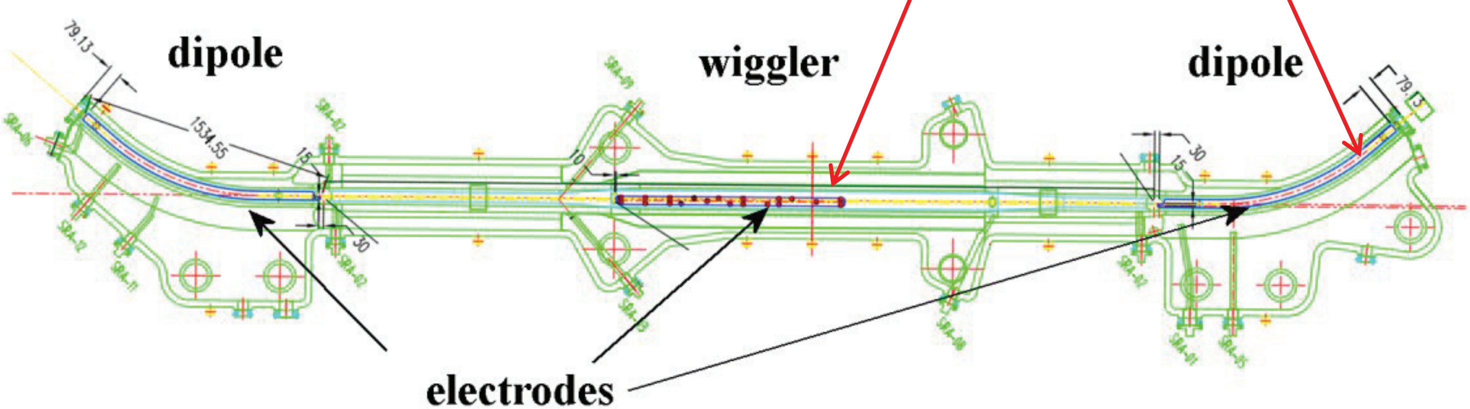
1. Anomalous pressure rise
2. Tune spread along bunch train
3. Fast horizontal multibunch instability
4. TMCI single bunch instability
5. others

Installation of Electrodes

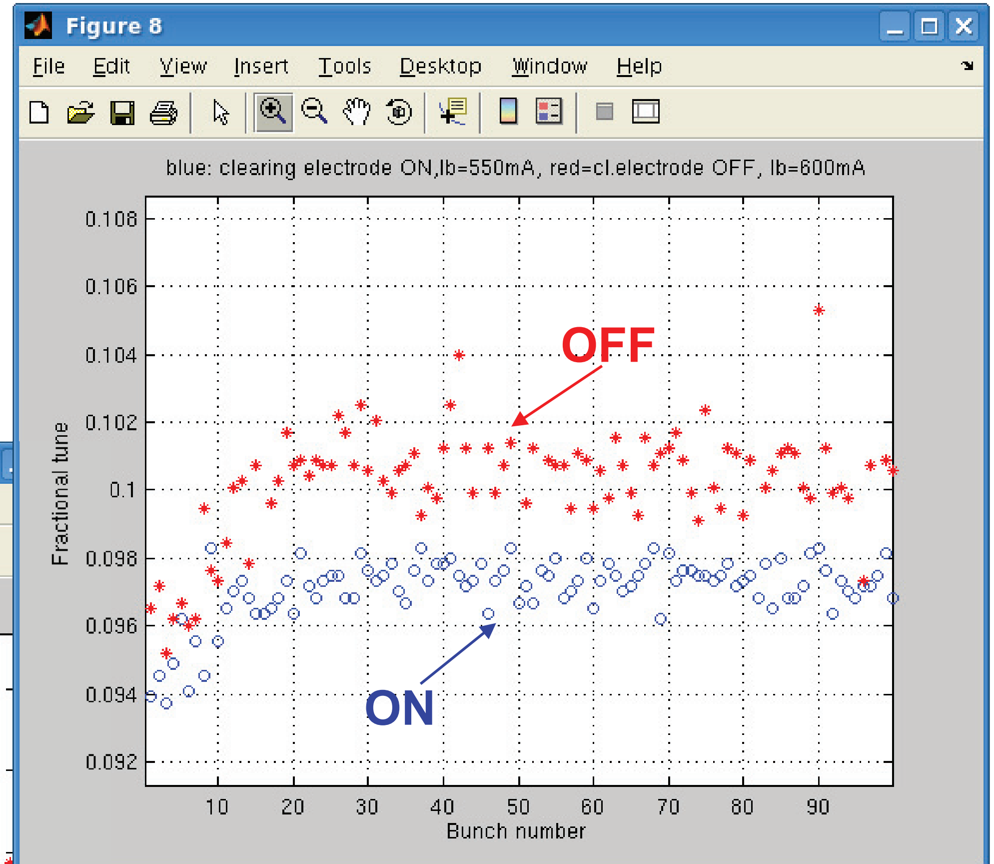
To mitigate the e-cloud instability *copper electrodes have been inserted in all dipole and wiggler chambers* of the machine and have been connected to external dc voltage generators.

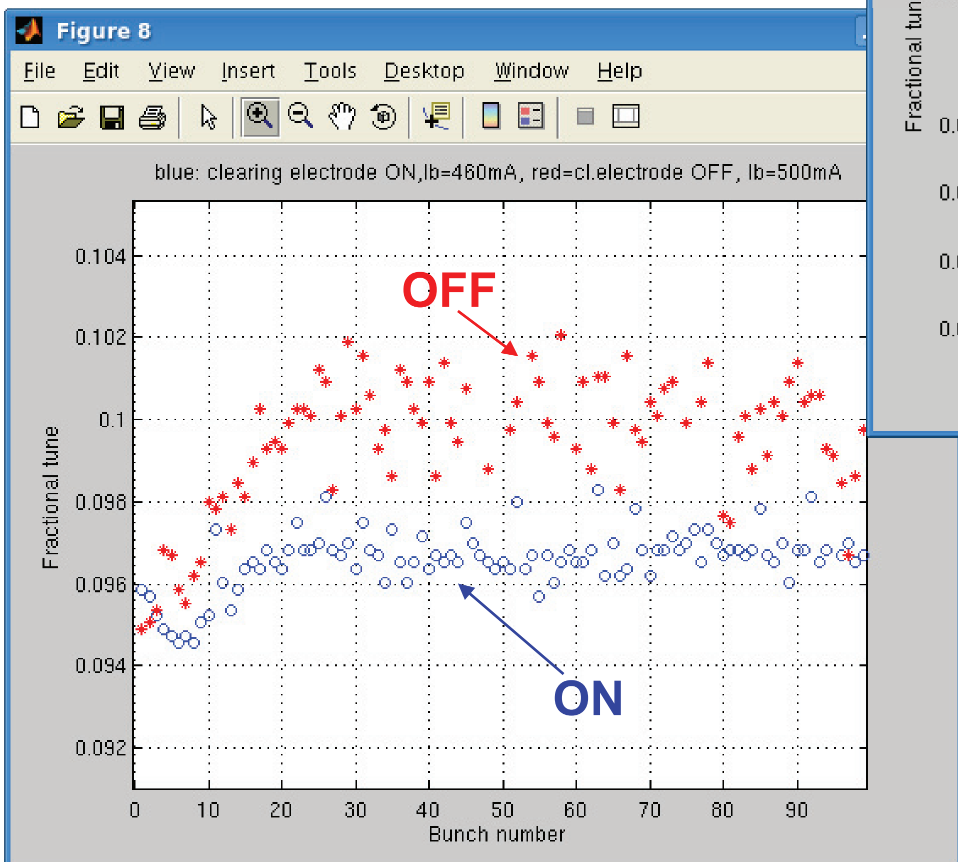
The dipole electrodes have a length of 1.4 or 1.6 m depending on the considered arc, while the wiggler ones are 1.4 m long.





Horizontal Bunch-by-Bunch Tune Spread Measured by the Feedback System



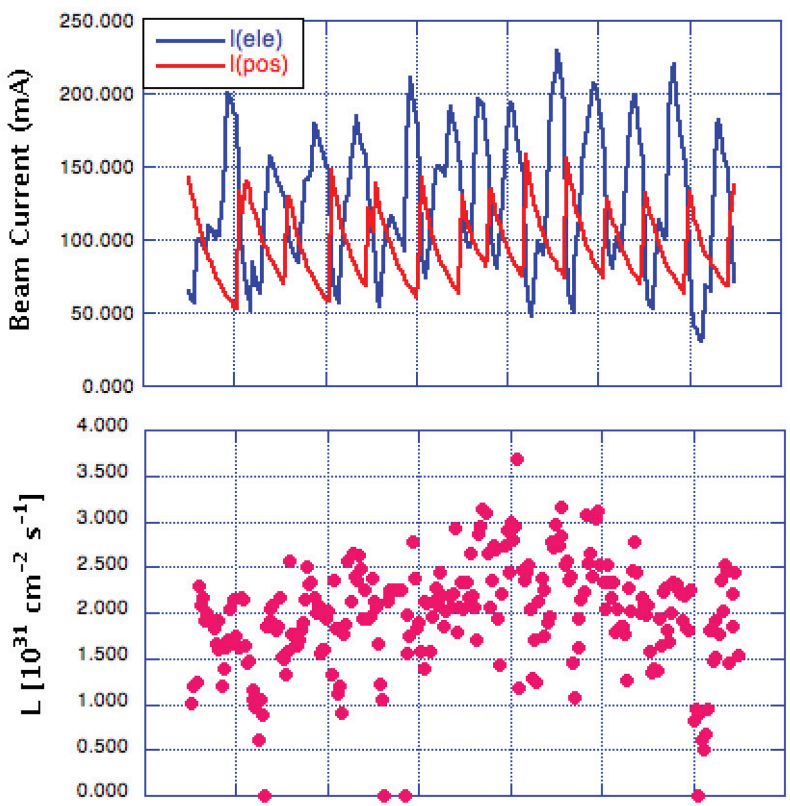


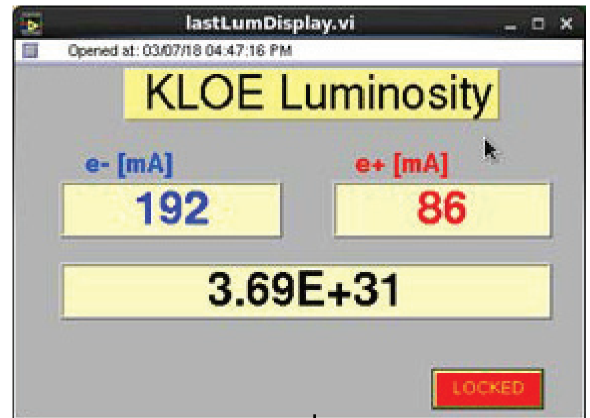
DAΦNE e+ beam:
100 bunches, spaced by 2.7ns with 20 buckets gap

Turning on the electrodes in 4 wigglers and 2 dipoles (not all) horizontal tune spread decreases

Collisions of 10 Bunches

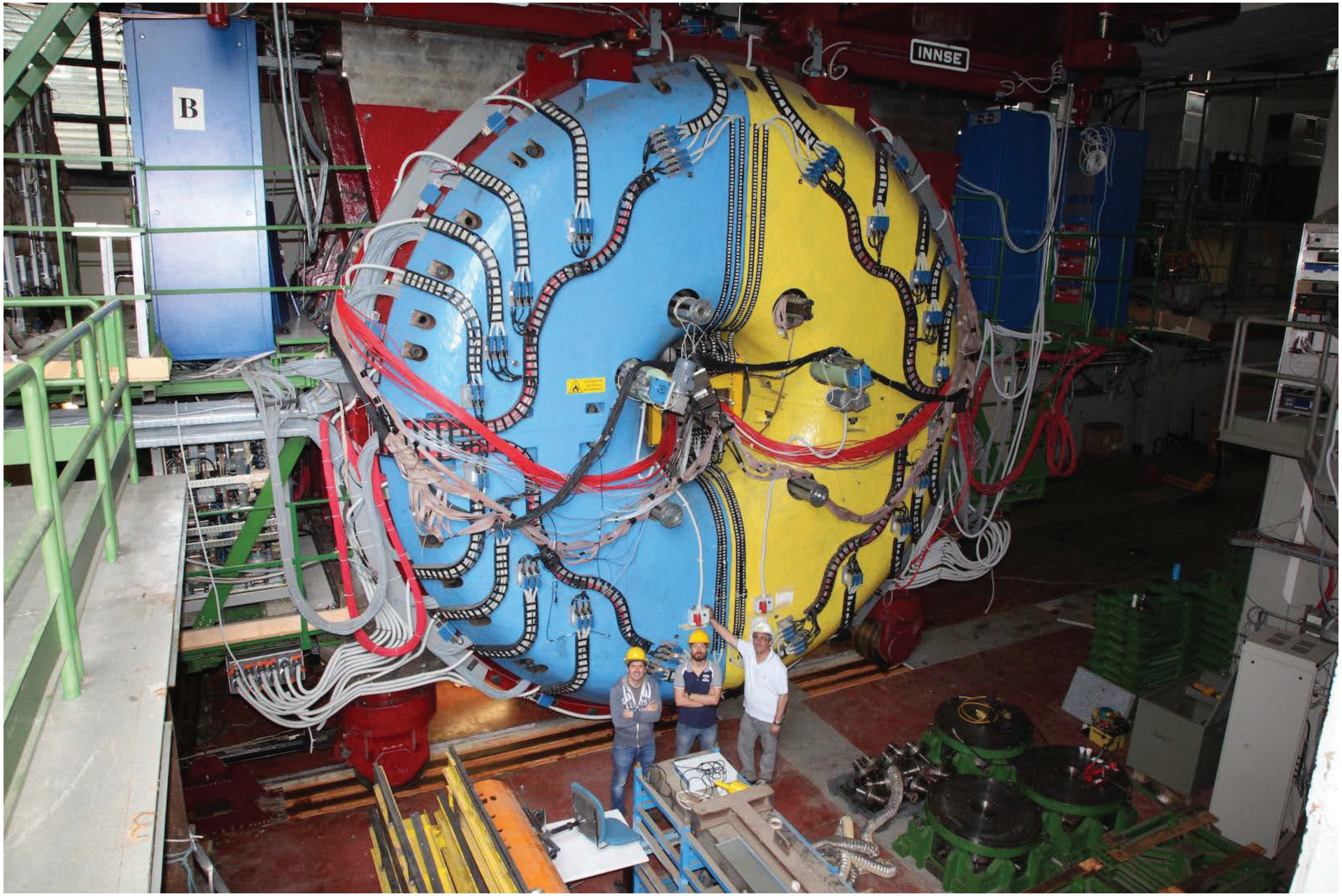
Experiment aimed at minimizing the impact of multi-bunch instabilities and e-cloud effects on the luminosity





1. The luminosty of $3.7x10^{32}$ $cm^{-2}s^{-1}$ might be potentially achieved colliding 100 bunches
2. Beam-beam is not a limiting factor
3. Crab Waist works

KLOE-2 leaves the stage

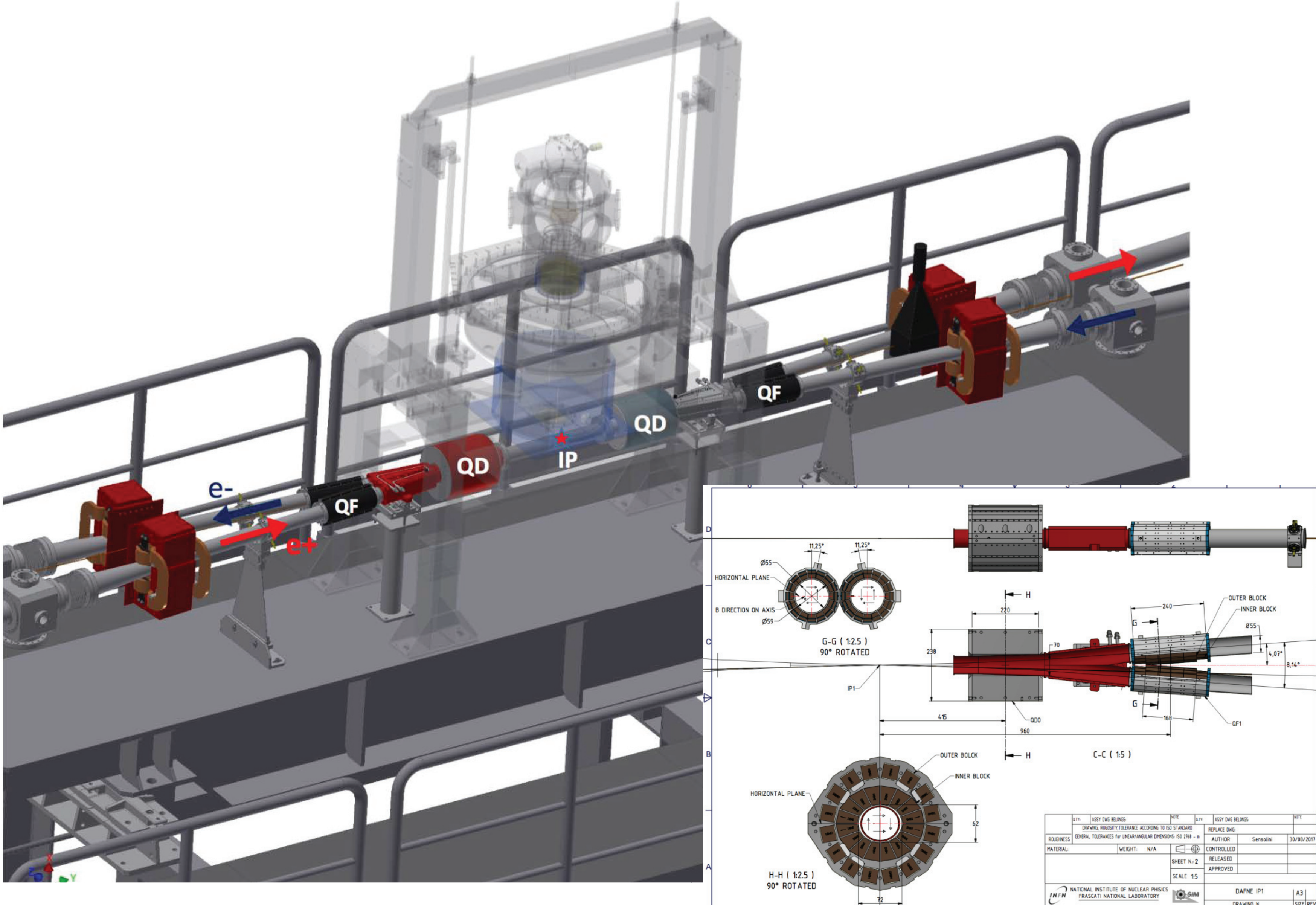


https://www.youtube.com/watch?v=ThDEwuhE1hA

Preparation for SIDDHARTA-2 Experimental Run

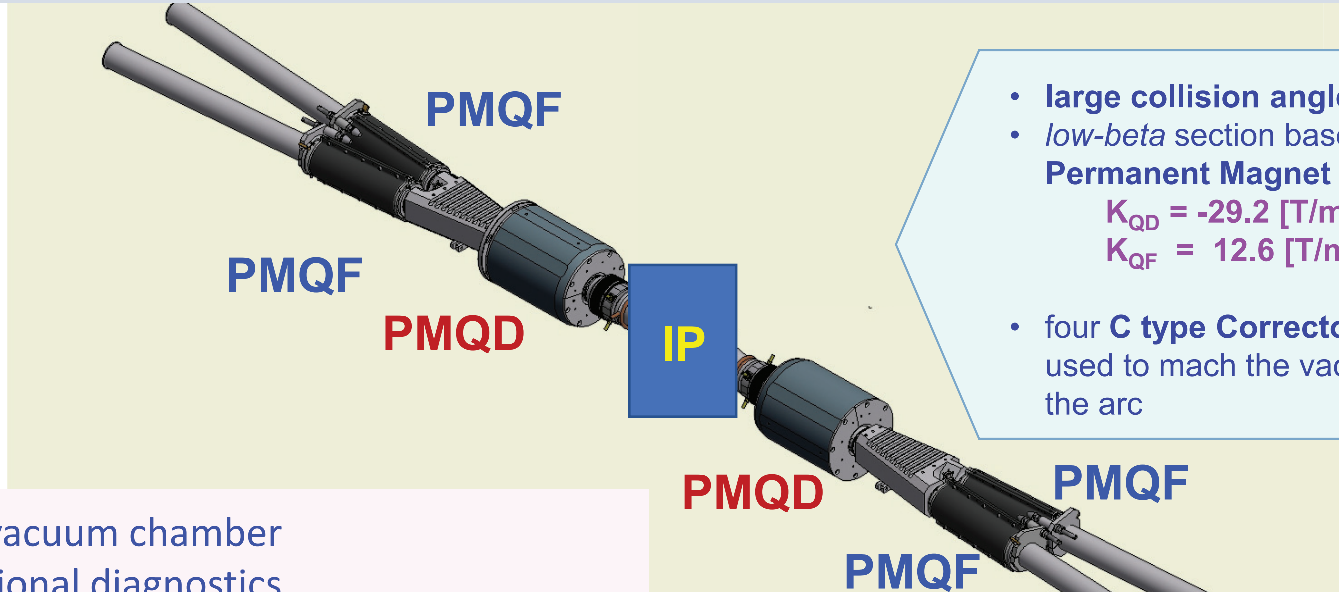
- New interaction region vacuum chamber
- New low-β insertion permanent magnet quadrupoles
- Installation of the second horizontal feedback system
- New fast absolute luminosity monitor adapted from the KLOE-2 experiment
- Implementation of the machine-experiment data exchange interface
- Collider general maintanence and consolidation

SIDDHARTA-2 IR



SIDDHARTA-2 low-β section

It is essentially based on the same design concepts used for the SIDDHARTA one



- **large collision angle ~ 50 mrd**
- *low-beta* section based on **Permanent Magnet QUADs**:
  K_{QD} **= -29.2 [T/m]**
  K_{QF} **= 12.6 [T/m]**

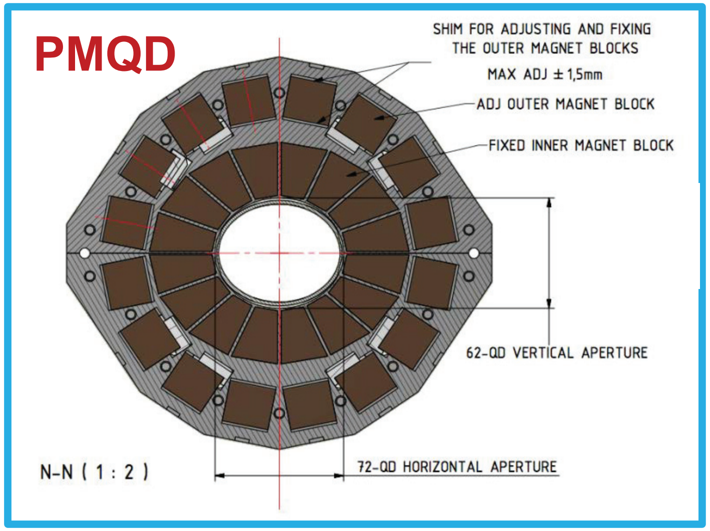
- four **C type Corrector** dipole are used to mach the vacuum chamber in the arc

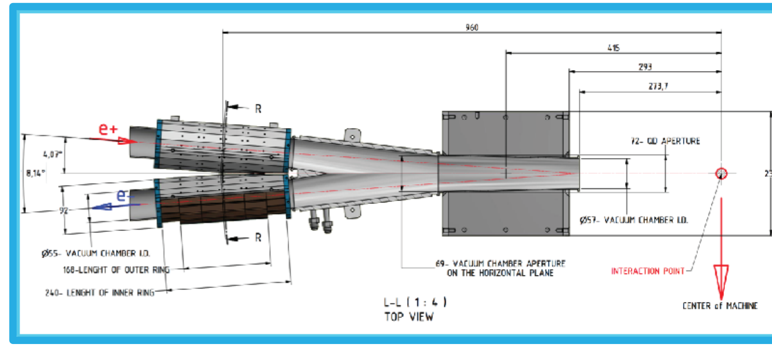
New vacuum chamber
Addidional diagnostics
The permanent magnet quadrupoles, that have been rebuilt aiming at:

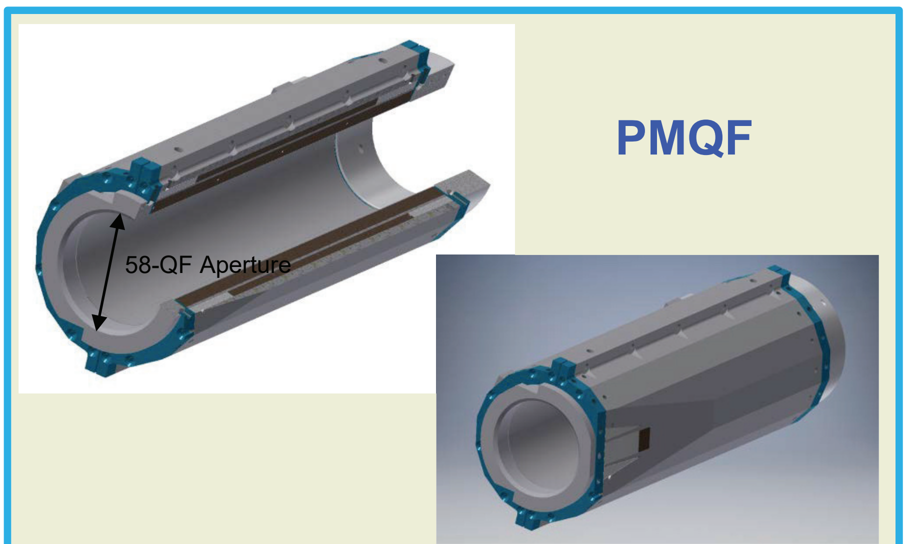
- speeding up the installation procedure
- improving design issues such as:
  - ❖ *good field region*
  - ❖ *gradient uniformity*
  - ❖ *mechanical assembly for PMQF*
  - ❖ ***aperture** for PMQD aiming at improving:*
    stay clear aperture
    background
    luminosity monitor efficiency

PMQs specifications

New PMQs are Halbach type magnets made of SmCo2:17
PMQs have been designed in collaboration with the ESRF magnet group.

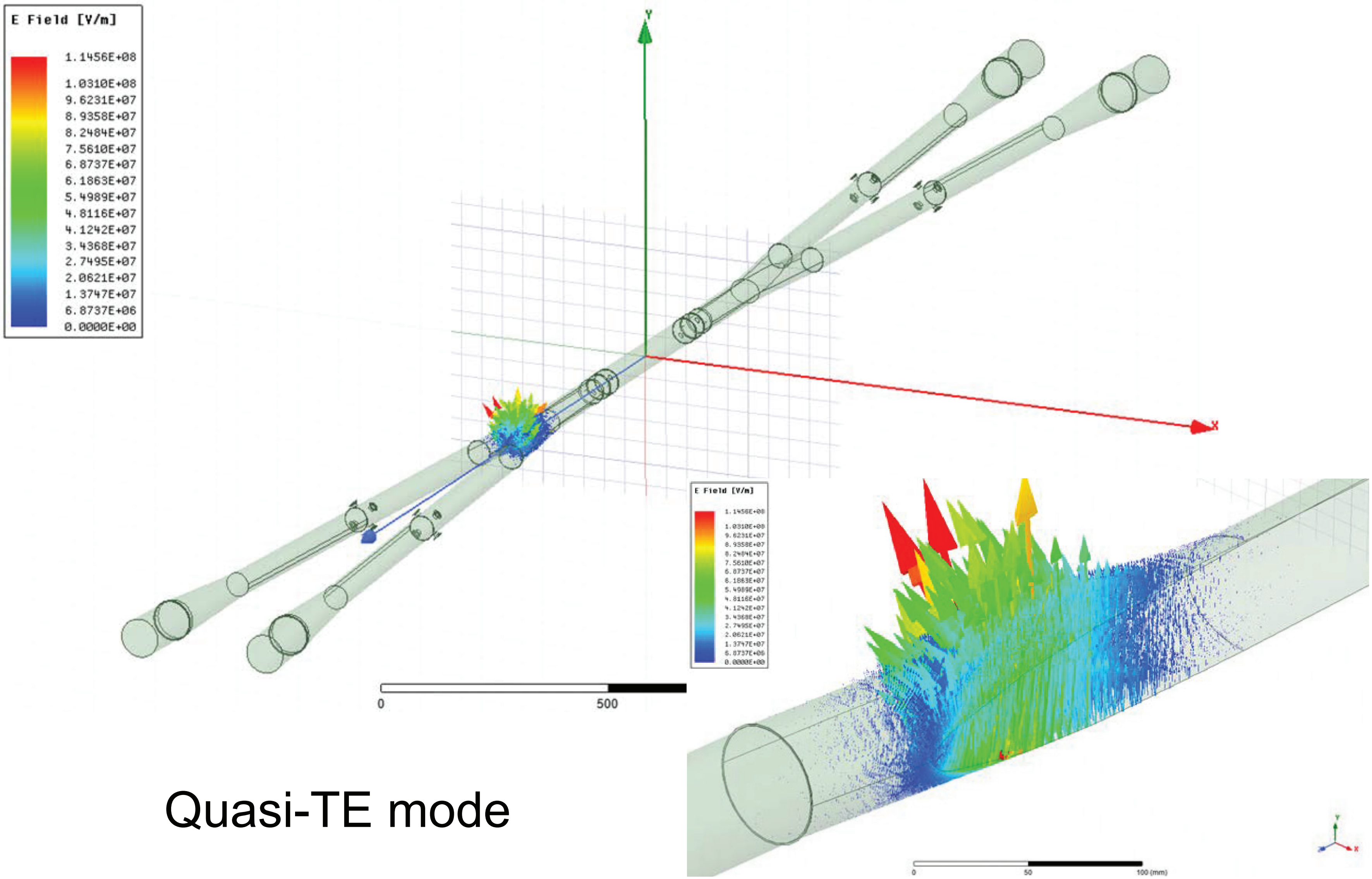






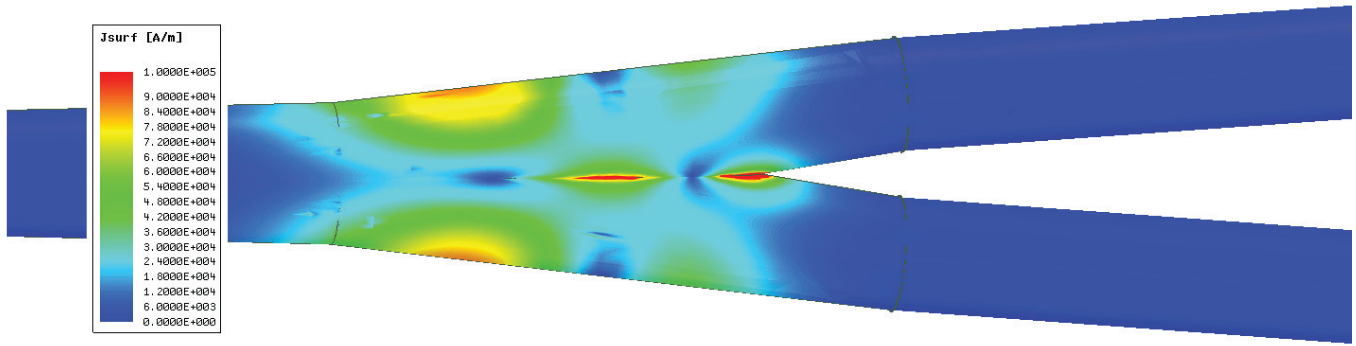
	PMQD	PMQF
Beam Pipe Aperture H-V (mm) at IP (I row) and at Y (II row) side	57 69 - 55	54
Inner Apert. With Case H-V (mm)	72 - 62	58
Outer Diameter H-V (mm)	238 - 220	95.6
Mech. Length Inner-Outer (mm)	220	168 - 240
Nominal Gradient (T/m)	29.2	12.6
Integrated Gradient (T)	6.7	3.0
Good Field Region (mm)	±20	±20
Integrated Field Quality |dB/B|	5.00E-4	5.00E-4
Magnet Assembly	2 halves	2 halves

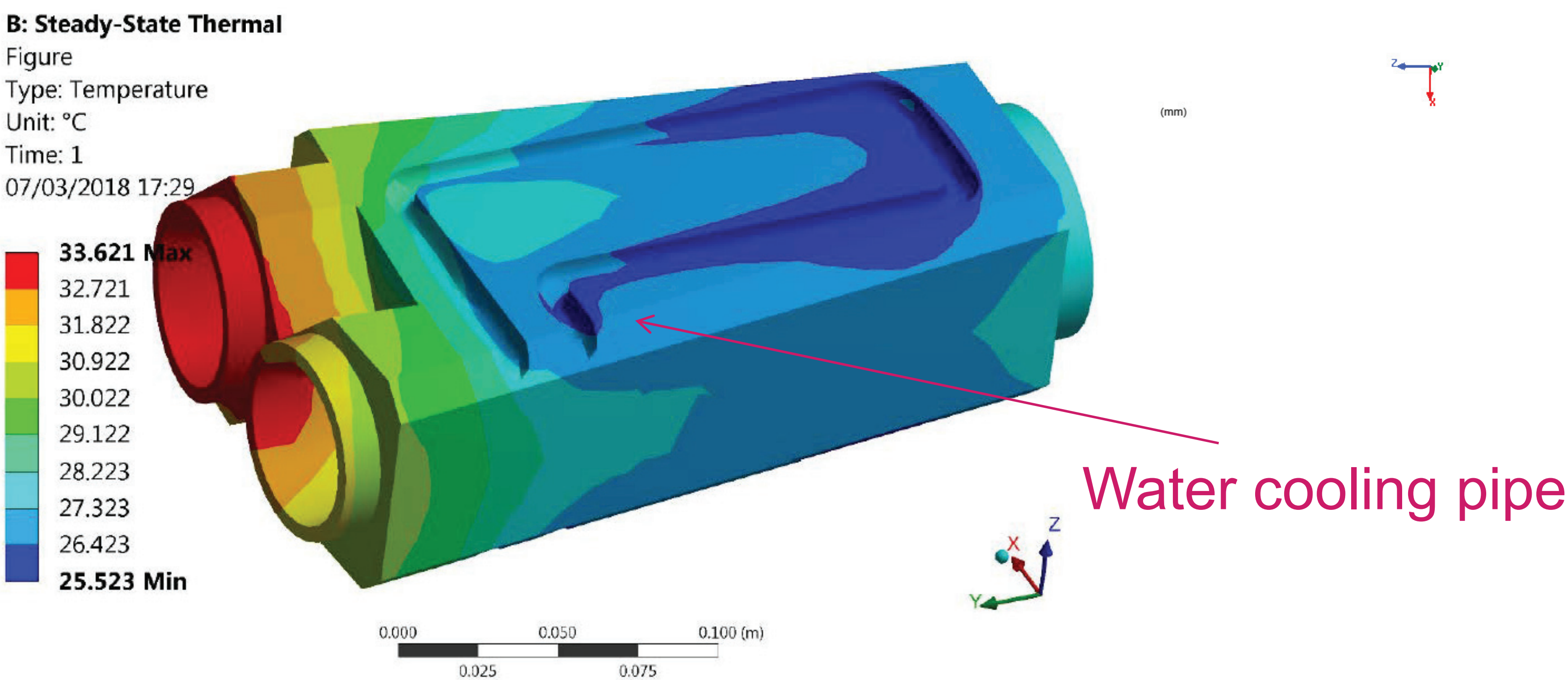
Mode 1 (and Mode 2 on the opposite side)



Quasi-TE mode

Temperature rise analysis





Conclusions

- DAΦNE has concluded the run for the KLOE-2 experiment achieving the high peak luminosity and exceeding the design integrated one. The Crab-Waist Collision Scheme has proven to be a viable approach to increase luminosity in circular colliders even in presence of an experimental apparatus strongly perturbing beam dynamics.

- An extensive work is in progress to prepare the run for the SIDDHARTA-2 experiment at DAΦNE. Several aspects of the collider and many subsystems have been upgraded in order to provide the highest performances in terms of luminosity and the lowest background contamination on the acquired data.

- The run for SIDDHARTA-2 will be the very last physics run of DAΦNE as a collider; thereafter the accelerator complex will most likely be converted into a test facility.